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# Symplectic Groupoids and Physical Applications

Bachelor's thesis  
in the field of Physics within MISMaP

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## Summary

This Bachelor's thesis develops an introductory route from Lie groupoids and Lie algebroids to Poisson geometry and symplectic groupoids, with emphasis on geometric mechanics and mathematical physics. We first introduce Lie groupoids and their infinitesimal counterparts, Lie algebroids, and then review Poisson manifolds, Casimir functions, symplectic leaves, Lie–Poisson dynamics, and the cotangent Lie algebroid of a Poisson manifold. We next study symplectic groupoids, the Poisson structure induced on their base, Lagrangian bisections, and the canonical identification of the Lie algebroid of a symplectic groupoid with the cotangent Lie algebroid of its base. The pair groupoid of a symplectic manifold and the cotangent groupoid  $T^*G \rightrightarrows \mathfrak{g}^*$  are used as the main examples. Finally, we apply the cotangent groupoid to the free rigid body. We derive the standard Lie–Poisson reduction, recover Euler's equations, and describe reconstruction and conservation of spatial angular momentum. The thesis is expository and aims to show how symplectic groupoids provide a global framework linking symplectic geometry, Poisson dynamics, and reduction by symmetry.

## Keywords

Lie groupoids, Lie algebroids, Poisson geometry, symplectic groupoids, Hamiltonian mechanics, Euler's equations

## Title of the thesis in Polish language

Grupoidy symplektyczne i zastosowania fizyczne

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# List of symbols

|                                                 |                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------|
| $M, N$                                          | : smooth finite-dimensional manifolds,                                                          |
| $C^\infty(M)$                                   | : smooth real functions on $M$ ,                                                                |
| $TM, T^*M$                                      | : tangent and cotangent bundles of $M$ ,                                                        |
| $\mathfrak{X}(M), \Omega^k(M)$                  | : vector fields and differential $k$ -forms on $M$ ,                                            |
| $\Gamma(E)$                                     | : sections of a vector bundle $E \rightarrow M$ ,                                               |
| $\iota_X, \mathcal{L}_X$                        | : interior product and Lie derivative along $X$ ,                                               |
| $\langle \cdot, \cdot \rangle$                  | : pairing of covectors with vectors, or of $\mathfrak{g}^*$ with $\mathfrak{g}$ ,               |
| $G, e$                                          | : Lie group and its identity element,                                                           |
| $\mathfrak{g}, \mathfrak{g}^*$                  | : Lie algebra of $G$ and its dual,                                                              |
| $L_g, R_g, C_g$                                 | : left translation, right translation, conjugation by $g$ ,                                     |
| $\text{Ad}_g, \text{Ad}_g^*$                    | : adjoint and coadjoint actions of $g \in G$ ,                                                  |
| $\text{ad}_\xi, \text{ad}_\xi^*$                | : adjoint and coadjoint operators of $\xi \in \mathfrak{g}$ ,                                   |
| $\xi^L$                                         | : left-invariant vector field on $G$ with $\xi^L(e) = \xi$ ,                                    |
| $\theta^L$                                      | : left Maurer–Cartan form, $\theta_g^L = T_g L_{g^{-1}}$ ,                                      |
| $\mathcal{O}_\mu$                               | : coadjoint orbit through $\mu \in \mathfrak{g}^*$ ,                                            |
| $\ell_\xi$                                      | : linear function $\mu \mapsto \langle \mu, \xi \rangle$ on $\mathfrak{g}^*$ ,                  |
| $d_\mu f$                                       | : differential of $f \in C^\infty(\mathfrak{g}^*)$ at $\mu$ , as an element of $\mathfrak{g}$ , |
| $a(\xi)$                                        | : fundamental vector field of $\xi$ for an action of $G$ ,                                      |
| $\hat{\xi}$                                     | : hat map $\mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ , $\hat{\xi}\eta = \xi \times \eta$ ,     |
| $\mathcal{G} \rightrightarrows M$               | : Lie groupoid with arrows $\mathcal{G}$ and base $M$ ,                                         |
| $s, t$                                          | : source and target maps,                                                                       |
| $m, u, \iota$                                   | : multiplication, unit map and inversion of a groupoid,                                         |
| $1_x$                                           | : unit arrow at $x \in M$ ,                                                                     |
| $\mathcal{G}^{(2)}$                             | : composable pairs $(g, h)$ , $s(g) = t(h)$ ,                                                   |
| $\mathcal{G}_x, \mathcal{G}^y, \mathcal{G}_x^y$ | : source fibre over $x$ , target fibre over $y$ , arrows $x \rightarrow y$ ,                    |
| $\mathcal{O}_x$                                 | : orbit of $x \in M$ under $\mathcal{G}$ ,                                                      |
| $\mathcal{G}_x^x$                               | : isotropy group at $x$ ,                                                                       |
| $L_g, R_g$                                      | : left and right translation by an arrow $g \in \mathcal{G}$ ,                                  |
| $M \times M$                                    | : pair groupoid of $M$ ,                                                                        |
| $G \ltimes M$                                   | : action groupoid of an action of $G$ on $M$ ,                                                  |

|                                                          |                                                                                                       |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| $A(\mathcal{G})$                                         | Lie algebroid of $\mathcal{G}$ , $A(\mathcal{G}) = \ker(Tt) _{u(M)}$ ,                                |
| $\rho$                                                   | anchor of a Lie algebroid,                                                                            |
| $[\cdot, \cdot]_A$                                       | bracket on the sections of a Lie algebroid $A$ ,                                                      |
| $\overleftarrow{\alpha}$ , $\mathfrak{X}^L(\mathcal{G})$ | left-invariant extension of $\alpha \in \Gamma(A(\mathcal{G}))$ ; all such fields,                    |
| $\mathfrak{g}_x$                                         | isotropy Lie algebra $\ker \rho_x$ ,                                                                  |
| $\mathfrak{g} \ltimes M$                                 | action algebroid,                                                                                     |
| $B$ , $\psi_B$                                           | bisection of $\mathcal{G}$ and its base map $t _B \circ (s _B)^{-1}$ ,                                |
| $(M, \omega)$ , $\omega^\flat$                           | symplectic manifold; $\omega^\flat: TM \rightarrow T^*M$ , $v \mapsto \iota_v \omega$ ,               |
| $\{\cdot, \cdot\}$                                       | Poisson bracket of functions,                                                                         |
| $\pi$                                                    | Poisson bivector, $\{f, g\} = \pi(df, dg)$ ,                                                          |
| $\pi^\sharp$                                             | bundle map $T^*M \rightarrow TM$ , $\langle \alpha, \pi^\sharp \beta \rangle = \pi(\alpha, \beta)$ ,  |
| $X_H$                                                    | Hamiltonian vector field, $X_H = \pi^\sharp dH = \{\cdot, H\}$ ,                                      |
| $\varphi_\tau$                                           | flow of $X_H$ on the base,                                                                            |
| $[\cdot, \cdot]_{\text{SN}}$                             | Schouten–Nijenhuis bracket,                                                                           |
| $\mathcal{D}$                                            | characteristic distribution $\text{im } \pi^\sharp$ ,                                                 |
| $L$ , $\omega_L$                                         | symplectic leaf and its symplectic form,                                                              |
| $C$                                                      | Casimir function,                                                                                     |
| $[\cdot, \cdot]_\pi$                                     | Koszul bracket of one-forms,                                                                          |
| $\{\cdot, \cdot\}_\pm$                                   | plus and minus Lie–Poisson brackets on $\mathfrak{g}^*$ ,                                             |
| $\overline{M}$                                           | Poisson manifold $(M, -\pi)$ ,                                                                        |
| $\Omega$                                                 | multiplicative symplectic form on a symplectic groupoid,                                              |
| $\overline{\mathcal{G}}$                                 | same manifold as $\mathcal{G}$ with symplectic form $-\Omega$ ,                                       |
| $\{\cdot, \cdot\}_\Omega$                                | Poisson bracket of $(\mathcal{G}, \Omega)$ ,                                                          |
| $W^\Omega$                                               | symplectic orthogonal complement of a subspace $W$ ,                                                  |
| $s^*f$ , $t^*f$                                          | pullbacks of $f \in C^\infty(M)$ to $\mathcal{G}$ ; $X_{s^*f}$ , $X_{t^*f}$ their Hamiltonian fields, |
| $\tilde{\varphi}_\tau$ , $\hat{\varphi}_\tau$            | flows of $X_{t^*H}$ and $X_{s^*H}$ on $\mathcal{G}$ ,                                                 |
| $B_\tau$                                                 | Lagrangian bisection swept out by the flow of $X_{s^*H}$ ,                                            |
| $\Psi$                                                   | isomorphism $A(\mathcal{G}) \rightarrow T^*M$ of Theorem 3.2,                                         |
| $\theta$                                                 | tautological one-form of $T^*G$ , $\Omega = -d\theta$ ,                                               |
| $(g, \mu)$                                               | arrow of $T^*G \rightrightarrows \mathfrak{g}^*$ in the left trivialisation,                          |
| $(\xi, \lambda)$                                         | tangent vector to $G \times \mathfrak{g}^*$ , $\xi = \theta_g^L(\dot{g})$ , $\lambda = \dot{\mu}$ ,   |
| $\hat{L}_h$                                              | cotangent lift of left translation, $(g, \mu) \mapsto (hg, \mu)$ ,                                    |
| $R$ , $\omega$                                           | attitude and body angular velocity of the rigid body,                                                 |
| $I$ , $I_1, I_2, I_3$                                    | inertia tensor and principal moments of inertia,                                                      |
| $\mu$                                                    | element of $\mathfrak{g}^*$ ; for the rigid body, the body angular momentum,                          |
| $J$                                                      | spatial angular momentum, $J = R\mu = t(R, \mu)$ ,                                                    |
| $\mu_i$ , $J_i$                                          | body and spatial momentum components, $\mu_i = s^*\ell_i$ , $J_i = t^*\ell_i$ ,                       |
| $b$ , $b_\tau$                                           | maps $\mathfrak{g}^* \rightarrow G$ describing bisections $\mu \mapsto (b(\mu), \mu)$ of $T^*G$ .     |

# Introduction

Symmetry and reduction are among the basic organising principles of classical mechanics [1, 14]. In the standard formulation a Lie group acts on a phase space and allows one to identify configurations or states that differ only by a symmetry. After reduction, however, the resulting phase space need not remain symplectic: it is naturally described by a Poisson structure, whose rank may vary and whose symplectic leaves carry the actual Hamiltonian dynamics. This passage from symplectic to Poisson geometry is one of the reasons why groupoids and algebroids arise naturally in geometric mechanics.

A groupoid may be viewed as a generalisation of a group in which a set of arrows between points in a manifold can be composed only when their source and target match. This apparently modest change is important geometrically: a group describes global symmetries, whereas a groupoid can encode local or partial equivalences between points. Lie groupoids add a compatible smooth structure, and their infinitesimal objects are Lie algebroids, in the same way that Lie algebras are the infinitesimal objects associated with Lie groups. These ideas provide a language in which local symmetry, reduction and global geometry can be treated in a common framework; standard references include [9, 17].

Poisson geometry provides the second ingredient. A Poisson manifold carries a Lie bracket on smooth functions which is also a derivation in each argument. Hamiltonian vector fields are therefore defined even when there is no globally non-degenerate symplectic form. The image of the Poisson tensor determines a possibly singular foliation by symplectic leaves, while its kernel is related to Casimir functions. For the dual  $\mathfrak{g}^*$  of a Lie algebra, the Lie–Poisson bracket gives one of the fundamental examples. Its symplectic leaves are the connected components of coadjoint orbits, and many reduced mechanical systems, including the free rigid body, live naturally on such spaces [6, 11].

The bridge between the infinitesimal Poisson picture and a global geometric object is supplied by symplectic groupoids. A symplectic groupoid is a Lie groupoid endowed with a multiplicative symplectic form. Multiplicativity is a strong compatibility condition. It makes the unit section Lagrangian, makes inversion anti-symplectic, and forces the source and target maps to encode opposite Poisson structures on the base. Moreover, the Lie algebroid of a symplectic groupoid is canonically identified with the cotangent Lie algebroid of its Poisson base. In this sense, a symplectic groupoid can be regarded as a global integration of Poisson geometry [16].

The aim of this Bachelor thesis is expository. It is not to develop a new theory of symplectic groupoids, but to present a self-contained path through the main constructions needed to understand why these objects are relevant to Hamiltonian mechanics for individual students of the University of Warsaw. Particular attention is paid to conventions and signs, because the source and target maps of the cotangent groupoid naturally produce opposite Lie–Poisson brackets. The pair groupoid of a symplectic manifold and the cotangent groupoid  $T^*G \rightrightarrows \mathfrak{g}^*$  serve as the main examples throughout the text.

This Bachelor thesis is organised as follows. Chapter 1 introduces groupoids, Lie groupoids

and Lie algebroids, together with their basic examples and the passage from a groupoid to its infinitesimal algebroid. Chapter 2 reviews Poisson geometry, Hamiltonian vector fields, Casimir functions and symplectic leaves; it then develops the Lie–Poisson structure on  $\mathfrak{g}^*$  and constructs the cotangent Lie algebroid of a Poisson manifold. Chapter 3 introduces symplectic groupoids and derives the structural consequences of multiplicativity, the induced Poisson structure on the base, Lagrangian bisections, and the identification of the groupoid Lie algebroid with the cotangent algebroid. Chapter 4 applies these constructions to  $T^*G \rightrightarrows \mathfrak{g}^*$  and, in particular, to the free rigid body. In that example the source records body angular momentum, the target records spatial angular momentum, Lie–Poisson reduction produces Euler’s equations, and reconstruction recovers the motion on  $T^*\mathrm{SO}(3)$ .

# Chapter 1

## Lie Groupoids and Lie Algebroids

### 1.1. From groups to groupoids

Groups are the standard tool to describe global symmetry in geometry [1]. A transitive group action describes a homogeneous space, which is by definition a manifold diffeomorphic to the quotient of a Lie group by a closed Lie subgroup [7, 8]. However, objects with repeated local patterns may have only a small global symmetry group. The following example of Weinstein [17] shows the problem and its solution.

Consider the tiling  $\mathcal{T}$  of the plane by  $2 \times 1$  rectangles, with edge set

$$\mathcal{T} = (2\mathbb{Z} \times \mathbb{R}) \cup (\mathbb{R} \times \mathbb{Z}) \subset \mathbb{R}^2.$$

Its symmetry group  $G_{\mathcal{T}}$ , the subgroup of the Euclidean group  $E(2)$  which preserves  $\mathcal{T}$ , contains the translations by the lattice  $2\mathbb{Z} \times \mathbb{Z}$  and several types of reflections. Consider now a finite region  $B = [0, 2m] \times [0, n]$ , where  $m, n$  are positive integers. The symmetry group of  $\mathcal{T} \cap B$  is just the Klein four-group generated by the two reflections through the centre of  $B$ , independently of  $m$  and  $n$ . It consists of the transformations that preserve the whole pattern but it says nothing about the local symmetries which relate individual points.

To record such information, instead of a transformation  $\varphi$  alone, we consider the set of the triples

$$\mathcal{G}_{\mathcal{T}} = \{ (y, \varphi, x) : \varphi \in G_{\mathcal{T}}, x, y \in \mathbb{R}^2, y = \varphi(x) \}.$$

Two triples  $(z, \psi, y')$  and  $(y, \varphi, x)$  can be composed to  $(z, \psi \circ \varphi, x)$  if and only if  $y' = y$ . Every point  $x$  carries an identity element  $(x, \text{id}, x)$ , and every triple has the inverse  $(x, \varphi^{-1}, y)$ . In a nutshell, a set with such a partially defined multiplication, with identities and with inverses is called a *groupoid*, and its elements are called *arrows*. After the restriction to  $B$ , the groupoid of triples relates the points of  $B$  which occupy the same position in their tiles (such points lie in the same *orbit*) and it still contains the symmetries which fix a given point (its *isotropy group*).

In short, groups describe global homogeneous symmetry, while groupoids generalize them by describing also local and partial symmetries. They reduce to groups when the base consists of one point. In the rest of the chapter we add a smooth structure to this notion and construct its infinitesimal counterpart.

### 1.2. Lie groupoids

Unless stated otherwise, manifolds in this thesis are assumed to be Hausdorff and second countable. When integrability is mentioned later, the standard theory may allow a non-Hausdorff arrow manifold while keeping the base manifold Hausdorff.

### 1.2.1. Definitions

**Definition 1.1** (Groupoid). A *groupoid* consists of two sets,  $\mathcal{G}$  (the *arrows*) and  $M$  (the *objects* or *base*), together with structure maps

- (i) *source* and *target*:  $s, t: \mathcal{G} \rightarrow M$ ;
- (ii) *multiplication*:  $m: \mathcal{G}^{(2)} \rightarrow \mathcal{G}$ ,  $(g, h) \mapsto gh$ , defined on the set of *composable pairs*

$$\mathcal{G}^{(2)} = \{(g, h) \in \mathcal{G} \times \mathcal{G} : s(g) = t(h)\};$$
- (iii) *unit*:  $u: M \rightarrow \mathcal{G}$ ,  $x \mapsto 1_x$ ;
- (iv) *inverse*:  $\iota: \mathcal{G} \rightarrow \mathcal{G}$ ,  $g \mapsto g^{-1}$ ;

satisfying, whenever the relevant products are defined, the compatibility and associativity identities

$$s(gh) = s(h), \quad t(gh) = t(g), \quad (gh)k = g(hk),$$

and the unit and inverse identities

$$s(1_x) = t(1_x) = x, \quad 1_{t(g)} g = g = g 1_{s(g)}, \quad g g^{-1} = 1_{t(g)}, \quad g^{-1} g = 1_{s(g)}.$$

We denote an arrow  $g \in \mathcal{G}$  with  $s(g) = x$  and  $t(g) = y$  as  $y \xleftarrow{g} x$ , and a groupoid as  $\mathcal{G} \rightrightarrows M$  to record the two maps  $s, t$  from arrows to objects. Two arrows compose precisely when the source of the first one matches the target of the second.

Equivalently, a groupoid is a small category in which every arrow is invertible [12, 17]. Recall that a small category is a category in which the space of objects and the space of morphisms are sets. If  $M = \{*\}$  is a single point, then every pair of arrows is composable, and the structure reduces to a group.

A groupoid in the sense of Definition 1.1 is sometimes called *set-theoretic*, to emphasise that no topology or smooth structure is assumed. If  $\mathcal{G}$  and  $M$  carry topologies and all structure maps are continuous, the groupoid is *topological*.

**Definition 1.2** (Lie groupoid). A *Lie groupoid* is a groupoid  $\mathcal{G} \rightrightarrows M$  in which  $\mathcal{G}$  and  $M$  are smooth manifolds, all structure maps are smooth, and the source and target maps  $s, t: \mathcal{G} \rightarrow M$  are surjective submersions [9, 12, 13].

The submersion condition on  $s$  and  $t$  has two consequences. First, the set  $\mathcal{G}^{(2)}$  of composable pairs is a smooth embedded submanifold of  $\mathcal{G} \times \mathcal{G}$ . Second, every source fibre  $s^{-1}(x)$  and every target fibre  $t^{-1}(x)$  is a closed embedded submanifold of  $\mathcal{G}$  [9].

*Remark 1.1.* Smoothness is a genuine additional requirement. Many groupoids that arise naturally are only topological. For this reason Definition 1.1 is kept alongside Definition 1.2.

### 1.2.2. First properties and notation

For  $x, y \in M$  write

$$\mathcal{G}_x = s^{-1}(x), \quad \mathcal{G}^y = t^{-1}(y), \quad \mathcal{G}_x^y = \mathcal{G}_x \cap \mathcal{G}^y$$

for the *source fibre*, the *target fibre*, and the set of arrows from  $x$  to  $y$ . The *orbit* of  $x \in M$  is

$$\mathcal{O}_x = t(\mathcal{G}_x) = \{y \in M : \mathcal{G}_x^y \neq \emptyset\},$$

the set of objects connected from  $x$  by an arrow. The orbits partition  $M$ . The *isotropy group* (or *vertex group*) at  $x$  is

$$\mathcal{G}_x^x = \{g \in \mathcal{G} : s(g) = t(g) = x\}.$$

It is a group under the restricted multiplication, and for a Lie groupoid it is a Lie group [9]. A Lie groupoid is *transitive* if it has a single orbit.

The unit map  $u: M \rightarrow \mathcal{G}$  is a smooth embedding:  $s \circ u = \text{id}_M$  shows that  $u$  is an injective immersion, and its image  $u(M)$ , which we identify with  $M$ , is a closed embedded submanifold of  $\mathcal{G}$  [9].

The analogues of left and right multiplication on a group are the *translations* between fibres. Fix  $g \in \mathcal{G}$  with  $s(g) = x$  and  $t(g) = y$ . Left multiplication by  $g$ , namely

$$L_g: \mathcal{G}^x \longrightarrow \mathcal{G}^y, \quad L_g(h) = gh,$$

is a diffeomorphism between target fibres, with inverse  $L_{g^{-1}}$ . Right multiplication  $R_g: \mathcal{G}_y \rightarrow \mathcal{G}_x$ ,  $R_g(h) = hg$ , is a diffeomorphism between source fibres. Left translations are used in Section 1.3 to construct the Lie algebroid of  $\mathcal{G}$ .

### 1.2.3. Examples

The following three Lie groupoids are used throughout the thesis.

**Example 1.1** (Lie group). A Lie group  $G$  is a Lie groupoid  $G \rightrightarrows *$  over a point. It has a single orbit, and its isotropy group is  $G$  itself.

**Example 1.2** (Pair groupoid). For any smooth manifold  $M$ , the product  $M \times M$  is a Lie groupoid over  $M$ , the *pair groupoid*, with

$$s(y, x) = x, \quad t(y, x) = y, \quad (z, y)(y, x) = (z, x), \quad u(x) = (x, x), \quad (y, x)^{-1} = (x, y).$$

Thus,  $(y, x)$  is the unique arrow from  $x$  to  $y$ . Source and target are the two projections, hence submersions, and the multiplication is smooth. The pair groupoid is transitive with trivial isotropy groups.

**Example 1.3** (Action groupoid). Let a Lie group  $G$  act smoothly on a manifold  $M$  from the left. The *action groupoid*  $G \ltimes M \rightrightarrows M$  has as arrows the pairs  $(g, x) \in G \times M$ , with

$$s(g, x) = x, \quad t(g, x) = g \cdot x, \quad (h, g \cdot x)(g, x) = (hg, x),$$

unit  $u(x) = (e, x)$  and inverse  $(g, x)^{-1} = (g^{-1}, g \cdot x)$ . The orbits and isotropy groups of  $G \ltimes M$  coincide with the orbits and stabilisers of the action. The groupoid of triples in Section 1.1 is the action groupoid of  $G_{\mathcal{T}}$  acting on  $\mathbb{R}^2$ .

The three examples display the two extreme cases and a mixture of them: the Lie group has a trivial base and full isotropy, the pair groupoid is transitive with trivial isotropy, and the action groupoid has, in general, non-trivial orbits and non-trivial isotropy groups.

## 1.3. From Lie groupoids to Lie algebroids

Every Lie group determines a Lie algebra by linearisation of the multiplication at the identity. This section makes a similar construction for a Lie groupoid. The resulting object is a *Lie algebroid* [9, 10]. We first construct it from a Lie groupoid and then provide the general definition.

### 1.3.1. Construction of the Lie algebroid

Recall first the group case. A vector field  $X \in \mathfrak{X}(G)$  on a Lie group  $G$  is *left-invariant* if  $(L_g)_*X = X$  for every  $g \in G$ . Since each  $(L_g)_*$  preserves the Lie bracket, the left-invariant vector fields  $\mathfrak{X}^L(G)$  form a Lie subalgebra of  $\mathfrak{X}(G)$ . Evaluation at the identity is

a linear isomorphism  $\mathfrak{X}^L(G) \rightarrow T_e G$  with inverse  $v \mapsto \overleftarrow{v}$  and  $\overleftarrow{v}_g = T_e L_g(v)$ . The bracket transported to  $T_e G$  defines the Lie algebra  $\mathfrak{g} = \text{Lie}(G)$ . Two features of this construction generalise: a left-invariant vector field is determined by its values at the identities, and its bracket is inherited from the bracket of vector fields. A Lie groupoid has an identity arrow  $1_x$  at every point  $x \in M$ , so the values at the identities form a section of a vector bundle over  $M$  rather than a single tangent vector.

Let  $\mathcal{G} \rightrightarrows M$  be a Lie groupoid. A vector field  $X \in \mathfrak{X}(\mathcal{G})$  is *left-invariant* if

- (i)  $X$  is tangent to the target fibres, namely  $T_g t(X_g) = 0$  for all  $g \in \mathcal{G}$ ; and
- (ii)  $X$  is equivariant under left translation, in other words  $T_h L_g(X_h) = X_{gh}$  for every  $(g, h) \in \mathcal{G}^{(2)}$ .

Condition (i) says that at every arrow  $g$  the vector  $X_g$  is tangent to the target fibre  $\mathcal{G}^{t(g)}$ . Left translations map target fibres to target fibres (Section 1.2), so condition (ii) makes sense, and it determines  $X$  on the whole  $\mathcal{G}$  from the values at the units.

The set  $\mathfrak{X}^L(\mathcal{G})$  of left-invariant vector fields is a Lie subalgebra of  $\mathfrak{X}(\mathcal{G})$ . Tangency to the target fibres is preserved by the bracket. This means that if  $X(f \circ t) = Y(f \circ t) = 0$  for all  $f \in C^\infty(M)$ , then  $[X, Y](f \circ t) = X(Y(f \circ t)) - Y(X(f \circ t)) = 0$ . Left-invariance is preserved because each  $L_g$  is a diffeomorphism between target fibres, so  $(L_g)_*[X, Y] = [(L_g)_*X, (L_g)_*Y] = [X, Y]$ .

Define the vector bundle

$$A(\mathcal{G}) := \ker(Tt)|_{u(M)} \longrightarrow M,$$

with fibre  $A(\mathcal{G})_x = \ker(T_{1_x} t)$ , the tangent vectors at the identity arrow  $1_x$  that are tangent to the target fibre  $\mathcal{G}^x$ . Such a vector is an infinitesimal deformation of the unit arrow  $1_x$  which keeps the target fixed. The map  $\Phi: \Gamma(A(\mathcal{G})) \rightarrow \mathfrak{X}^L(\mathcal{G})$  given by

$$\Phi(\alpha)_g = \overleftarrow{\alpha}_g := T_{1_{s(g)}} L_g(\alpha_{s(g)}),$$

is an isomorphism with inverse  $\Phi^{-1}(X) = X|_{u(M)}$ . Transporting the bracket of left-invariant vector fields through  $\Phi$  defines a Lie bracket on  $\Gamma(A(\mathcal{G}))$  of the form

$$[\alpha, \beta] := \Phi^{-1}([\overleftarrow{\alpha}, \overleftarrow{\beta}]).$$

We call  $A(\mathcal{G})$  the *Lie algebroid of the Lie groupoid  $\mathcal{G}$* .

### 1.3.2. The anchor and the Leibniz rule

A left-invariant vector field is tangent to the target fibres but, in general, not to the source fibres. Applying  $Ts$  produces a vector field on the base. Define the *anchor*  $\rho: A(\mathcal{G}) \rightarrow TM$  by

$$\rho(\alpha_x) = T_{1_x} s(\alpha_x) \in T_x M, \quad \alpha_x \in A(\mathcal{G})_x = \ker(T_{1_x} t).$$

The anchor measures the displacement of the source. In other words,  $\rho(\alpha)$  is the vector field induced on  $M$  by the flow of  $\overleftarrow{\alpha}$ . Elements with  $\rho(\alpha_x) = 0$  move neither the source nor the target. They form the *isotropy Lie algebra*  $\mathfrak{g}_x = \ker \rho_x$ , which is the Lie algebra of the vertex group  $\mathcal{G}_x^x$ .

The bracket on  $\Gamma(A(\mathcal{G}))$  is  $\mathbb{R}$ -bilinear but not  $C^\infty(M)$ -bilinear: for all  $\alpha, \beta \in \Gamma(A(\mathcal{G}))$  and  $f \in C^\infty(M)$ ,

$$[\alpha, f\beta] = f[\alpha, \beta] + \rho(\alpha)(f)\beta. \tag{1.3.1}$$

To prove (1.3.1), we note first that the left-invariant extension of  $f\beta$  is  $(f \circ s) \overleftarrow{\beta}$ , where  $f \circ s$  is the pull-back of  $f$  to  $\mathcal{G}$  along the source map. The Leibniz rule for vector fields on  $\mathcal{G}$  gives

$$[\overleftarrow{\alpha}, (f \circ s) \overleftarrow{\beta}] = (f \circ s) [\overleftarrow{\alpha}, \overleftarrow{\beta}] + \overleftarrow{\alpha}(f \circ s) \overleftarrow{\beta}.$$

Now,  $\overleftarrow{\alpha}$  is  $s$ -related to  $\rho(\alpha)$ , so  $\overleftarrow{\alpha}(f \circ s) = \rho(\alpha)(f) \circ s$ , and the restriction to the unit section gives (1.3.1). The correction term  $\rho(\alpha)(f) \beta$  vanishes when  $\rho = 0$  or when  $f$  is constant. Because of this term, the operator  $[\alpha, \cdot]$  is a first-order differential operator. The value of  $[\alpha, \beta]$  at  $x$  depends on  $\beta(x)$  and also on the derivative of  $\beta$  in the direction  $\rho(\alpha)_x$ .

### 1.3.3. The abstract definition

**Definition 1.3** (Lie algebroid). A *Lie algebroid* over a manifold  $M$  is a vector bundle  $A \rightarrow M$  equipped with a Lie bracket  $[\cdot, \cdot]_A$  on  $\Gamma(A)$  and a vector bundle morphism  $\rho: A \rightarrow TM$ , called the *anchor*, satisfying the Leibniz rule

$$[\alpha, f\beta]_A = f[\alpha, \beta]_A + \rho(\alpha)(f) \beta \quad \text{for all } \alpha, \beta \in \Gamma(A), f \in C^\infty(M).$$

By construction,  $A(\mathcal{G})$  is a Lie algebroid over  $M$ . The compatibility of the anchor with the two brackets does not have to be assumed, since it follows from the definition.

**Proposition 1.1.** *In any Lie algebroid  $A \rightarrow M$  the anchor preserves brackets, i.e.*

$$\rho([\alpha, \beta]_A) = [\rho(\alpha), \rho(\beta)] \quad \text{for all } \alpha, \beta \in \Gamma(A).$$

*Proof.* Let  $\alpha, \beta, \gamma \in \Gamma(A)$  and  $f \in C^\infty(M)$ . We expand  $[\alpha, [\beta, f\gamma]_A]_A$  in two ways. First we apply the Leibniz rule twice directly. Then we use the Jacobi identity to write the same expression as  $[[\alpha, \beta]_A, f\gamma]_A + [\beta, [\alpha, f\gamma]_A]_A$  and we apply the Leibniz rule to each term. We compare the two expansions. The terms in which  $f$  multiplies a bracket of sections agree by the Jacobi identity for  $[\cdot, \cdot]_A$ , and the terms with a single derivative of  $f$  agree term by term. The remaining equality is the anchor defect applied to  $f$  and multiplied by  $\gamma$ . Pointwise, on a component of positive bundle rank one may choose  $\gamma$  nonzero at the point and cancel its value; on a rank-zero component both anchors vanish. Thus

$$\rho(\alpha)(\rho(\beta)(f)) = \rho([\alpha, \beta]_A)(f) + \rho(\beta)(\rho(\alpha)(f)).$$

Hence,  $\rho([\alpha, \beta]_A)(f) = [\rho(\alpha), \rho(\beta)](f)$  for every  $f$ , which proves the claim.  $\square$

*Remark 1.2.* The anchor is a morphism from  $(\Gamma(A), [\cdot, \cdot]_A)$  to  $(\mathfrak{X}(M), [\cdot, \cdot])$ . Its image  $\text{im } \rho \subseteq TM$  is an involutive distribution, perhaps of varying rank, whose integral submanifolds are the *orbits* of the algebroid. Its kernel  $\mathfrak{g}_x = \ker \rho_x$  at each point is a Lie algebra, the *isotropy Lie algebra*. For  $A = A(\mathcal{G})$  the isotropy Lie algebras agree with those of the groupoid. If the source fibres of  $\mathcal{G}$  are connected, then the algebroid and groupoid orbits agree. In general, the algebroid sees the orbits generated by the source-connected component of the groupoid.

*Remark 1.3.* A Lie algebroid is called *integrable* if it is isomorphic to  $A(\mathcal{G})$  for some Lie groupoid  $\mathcal{G} \rightrightarrows M$ . In contrast to the theory of Lie groups and Lie algebras, not every Lie algebroid is integrable. Namely, there is a genuine obstruction [5]. The general theory is not developed here. We use its consequences for cotangent algebroids and for the integration of Poisson manifolds in Chapter 3.

### 1.3.4. Examples

We compute the Lie algebroids of the three groupoids of Section 1.2.3.

**Example 1.4** (Lie algebra). A finite-dimensional Lie algebra  $\mathfrak{g}$  is a Lie algebroid over a point, with zero anchor and its own bracket. It is the Lie algebroid of any Lie group integrating it (Example 1.1). Since  $TM = 0$ , the correction term in the Leibniz rule vanishes.

**Example 1.5** (Tangent bundle). The tangent bundle  $TM \rightarrow M$ , with the Lie bracket of vector fields and with the anchor  $\rho = \text{id}_{TM}$ , is a Lie algebroid. It is the Lie algebroid of the pair groupoid  $M \times M \rightrightarrows M$  (Example 1.2). Indeed, for  $t(y, x) = y$  the target fibre over  $x$  is  $\{x\} \times M$ , and its tangent space at the unit  $(x, x)$  is identified with  $T_x M$ , so  $A(M \times M) \cong TM$ . The anchor  $\rho = Ts$  is the identity map, because  $s$  is the projection onto the second factor. The kernel  $\ker \rho = 0$  corresponds to the trivial isotropy of the pair groupoid.

**Example 1.6** (Action algebroid). Let a Lie group  $G$  act on  $M$  from the left, with fundamental vector fields  $a: \mathfrak{g} \rightarrow \mathfrak{X}(M)$ ,  $a(\xi)_x = \frac{d}{d\varepsilon}\big|_0 \exp(\varepsilon\xi) \cdot x$ . We compute the Lie algebroid of the action groupoid  $G \ltimes M \rightrightarrows M$  of Example 1.3.

*The bundle and the anchor.* Differentiate  $t(g, x) = g \cdot x$  at the unit  $(e, x)$  along a curve  $\varepsilon \mapsto (\exp(\varepsilon\xi), x_\varepsilon)$  with velocity  $(\xi, v)$ . This gives

$$T_{(e,x)}t(\xi, v) = a(\xi)_x + v.$$

Hence, the fibre of the Lie algebroid is

$$A(G \ltimes M)_x = \ker T_{(e,x)}t = \{(\xi, -a(\xi)_x) : \xi \in \mathfrak{g}\},$$

and the first component identifies it with  $\mathfrak{g}$ . Therefore the Lie algebroid is the trivial bundle  $A = M \times \mathfrak{g}$ , called the *action algebroid*  $\mathfrak{g} \ltimes M$ . Its sections are the  $\mathfrak{g}$ -valued functions  $\sigma \in C^\infty(M, \mathfrak{g})$ . The anchor is  $Ts$ , and  $s(g, x) = x$  returns the second component. This shows that on constant sections

$$\rho(\xi)_x = -a(\xi)_x.$$

*Left-invariant extensions.* Left translation by  $(g, h \cdot x)$  sends  $(h, x)$  to  $(gh, x)$ . Consequently, its differential sends  $(w, v)$  to  $(T_h L_g w, v)$ . By the construction of Section 1.3.1 the left-invariant extension of the constant section  $\xi$  is

$$\overleftarrow{\xi}(g, x) = (T_e L_g \xi, -a(\xi)_x) = (\xi_g^L, -a(\xi)_x).$$

The two components separate. The first is the left-invariant field  $\xi^L$  on  $G$ , which does not depend on  $x$ . The second is the field  $-a(\xi)$  on  $M$ , which does not depend on  $g$ .

*The bracket on constant sections.* Vector fields of this separated form bracket componentwise, because each component is constant in the other variable. It follows that

$$[\overleftarrow{\xi}, \overleftarrow{\eta}] = ([\xi^L, \eta^L], [-a(\xi), -a(\eta)]) = ([\xi, \eta]^L, [a(\xi), a(\eta)]).$$

By Section 1.3.1 the bracket of two left-invariant fields is again left-invariant, so its value at the unit  $(e, x)$  lies in the fibre  $A(G \ltimes M)_x$ . The elements of that fibre are the pairs  $(\zeta, -a(\zeta)_x)$ , and the first component of the value is  $[\xi, \eta]$ . This pins the value,

$$[\overleftarrow{\xi}, \overleftarrow{\eta}](e, x) = ([\xi, \eta], -a([\xi, \eta])_x).$$

Hence, on constant sections  $[\xi, \eta]_A = [\xi, \eta]$ . Comparing the second components yields the identity

$$[a(\xi), a(\eta)] = -a([\xi, \eta]).$$

With the convention for fundamental vector fields used here, a left action reverses brackets. The reversal is also what makes the anchor a homomorphism, since  $\rho([\xi, \eta]) = -a([\xi, \eta]) = [a(\xi), a(\eta)] = [\rho(\xi), \rho(\eta)]$ , as Proposition 1.1 requires.

*The bracket on all sections.* Fix a basis  $\xi_1, \dots, \xi_n$  of  $\mathfrak{g}$  and write  $\sigma = \sigma^i \xi_i$  and  $\tau = \tau^j \xi_j$  with coefficients in  $C^\infty(M)$ . The Leibniz rule (1.3.1), its mirror through antisymmetry, and  $\sigma^i \rho(\xi_i) = \rho(\sigma)$  give

$$[\sigma, \tau]_A = \sigma^i \tau^j [\xi_i, \xi_j] + \rho(\sigma)(\tau^j) \xi_j - \rho(\tau)(\sigma^i) \xi_i = [\sigma, \tau]_{\mathfrak{g}} - a(\sigma) \tau + a(\tau) \sigma.$$

Here,  $[\sigma, \tau]_{\mathfrak{g}}$  is the pointwise bracket, and  $a(\sigma) \tau$  is the derivative of the  $\mathfrak{g}$ -valued function  $\tau$  along the vector field  $x \mapsto a(\sigma(x))_x$ . Neither side depends on the choice of basis.

## Chapter 2

# Poisson Manifolds and Cotangent Lie Algebroids

This chapter develops the Poisson geometry used in the rest of the thesis. It has two deliverables. The first is the *symplectic foliation* of a Poisson manifold, together with its Casimir functions. The second is the *cotangent Lie algebroid*  $T^*M$ , whose relation to symplectic groupoids is studied in Chapter 3. The dual  $\mathfrak{g}^*$  of a Lie algebra runs through both as the worked example, and it is where the free rigid body lives.

Section 2.1 defines Poisson manifolds through the properties of the symplectic bracket that survive without non-degeneracy. Section 2.2 treats Hamiltonian dynamics, Casimir functions and the decomposition into symplectic leaves. Section 2.3 studies the Lie–Poisson structure on  $\mathfrak{g}^*$ , its coadjoint orbits and the free rigid body. Section 2.4 constructs the cotangent Lie algebroid.

The material is standard and is treated in the books of Vaisman [15], Crainic, Fernandes and Mărcuț [6], Cannas da Silva and Weinstein [3] and, for the mechanics, Marsden and Ratiu [11]. More precise references are given at each step.

### 2.1. From symplectic to Poisson manifolds

The simplest models of conservative mechanical systems have symplectic phase spaces. Their defining 2-form is non-degenerate. After reduction by symmetries, a smooth quotient may instead carry a degenerate Poisson structure; constrained systems require additional hypotheses and are not considered here. This section recalls the symplectic bracket and then uses the properties that do not rely on non-degeneracy to define a Poisson manifold.

Let  $(M, \omega)$  be a symplectic manifold. Non-degeneracy means that the bundle map

$$\omega^\flat: TM \longrightarrow T^*M, \quad v \longmapsto \iota_v \omega,$$

is an isomorphism. Each  $H \in C^\infty(M)$  then has a unique *Hamiltonian vector field*  $X_H$  defined by

$$\iota_{X_H} \omega = dH, \quad \text{that is} \quad X_H = (\omega^\flat)^{-1}(dH).$$

The trajectories of the system are its integral curves. The *Poisson bracket* of two functions is

$$\{f, g\} = \omega(X_f, X_g),$$

and a short computation gives  $X_H(f) = \{f, H\}$ . Therefore every observable evolves along the flow by  $\dot{f} = \{f, H\}$ .

This bracket has four properties. It is bilinear and antisymmetric. It satisfies the *Leibniz rule*

$$\{f, gh\} = \{f, g\}h + g\{f, h\},$$

because a Hamiltonian vector field is a derivation of  $C^\infty(M)$ . And it satisfies the *Jacobi identity*

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0,$$

which is the only one that uses  $d\omega = 0$ . Keeping the four properties without mentioning  $\omega$  gives the general notion.

**Definition 2.1** (Poisson manifold). A *Poisson manifold* is a smooth manifold  $M$  with a bilinear map

$$\{\cdot, \cdot\}: C^\infty(M) \times C^\infty(M) \longrightarrow C^\infty(M)$$

that is antisymmetric, satisfies the Jacobi identity, and is a derivation in each argument,

$$\{f, gh\} = \{f, g\}h + g\{f, h\} \quad \text{for all } f, g, h \in C^\infty(M).$$

The map  $\{\cdot, \cdot\}$  is called the *Poisson bracket* [6, 15].

Bilinearity, antisymmetry and the Jacobi identity make  $C^\infty(M)$  a Lie algebra; the Leibniz rule is the additional compatibility with multiplication of functions. As in the symplectic case, each  $H \in C^\infty(M)$  has a *Hamiltonian vector field*

$$X_H = \{\cdot, H\}.$$

The Leibniz rule makes  $\{\cdot, H\}$  a derivation of  $C^\infty(M)$ ; observables again evolve by  $\dot{f} = \{f, H\}$ . On a connected symplectic manifold, a Hamiltonian vector field determines its Hamiltonian only up to an additive constant. On a Poisson manifold, the ambiguity is larger: one may add any Casimir function, which need not be constant. As in the symplectic case, not every vector field is Hamiltonian.

We now record a bivector form of the bracket.

*Remark 2.1.* A map  $C^\infty(M)^k \rightarrow C^\infty(M)$  that is  $\mathbb{R}$ -multilinear and a derivation in each argument (a *k-derivation*) has, in any chart,

$$D(f_1, \dots, f_k) = \sum_{i_1, \dots, i_k} D(x^{i_1}, \dots, x^{i_k}) \partial_{i_1} f_1 \cdots \partial_{i_k} f_k.$$

This shows that its value at  $x$  depends only on the differentials there, and  $D \equiv 0$  as soon as it vanishes on coordinate functions. An antisymmetric 2-derivation is therefore  $\{f, g\} = \pi(df, dg)$  for a unique bivector field  $\pi \in \Gamma(\wedge^2 TM)$  with  $\pi^{ij} = \{x^i, x^j\}$  [6].

By Remark 2.1 the value  $\{f, g\}(x)$  depends only on  $df_x$  and  $dg_x$ . It is an antisymmetric 2-derivation, so there is a unique bivector field  $\pi \in \Gamma(\wedge^2 TM)$ , the *Poisson bivector*, with

$$\{f, g\} = \pi(df, dg) \quad \text{for all } f, g \in C^\infty(M).$$

The bivector defines a bundle map

$$\pi^\sharp: T^*M \longrightarrow TM, \quad \langle \alpha, \pi^\sharp \beta \rangle = \pi(\alpha, \beta),$$

the Poisson analogue of  $(\omega^\flat)^{-1}$ . The Hamiltonian vector field is

$$X_H = \pi^\sharp(dH),$$

because  $\langle df, \pi^\sharp dH \rangle = \pi(df, dH) = \{f, H\} = X_H(f)$ .

Of the four axioms, bilinearity, antisymmetry and the Leibniz rule hold for any bivector field. Only the Jacobi identity is a real condition on  $\pi$ . That condition is

$$[\pi, \pi]_{\text{SN}} = 0,$$

where  $[\cdot, \cdot]_{\text{SN}}$  is the Schouten–Nijenhuis bracket, the extension of the Lie bracket of vector fields to multivector fields [15]. We will not need this identity and state it only to record the conclusion: *a Poisson structure on  $M$  is the same datum as a bivector field  $\pi$  with  $[\pi, \pi]_{\text{SN}} = 0$ , the bracket being recovered by  $\{f, g\} = \pi(df, dg)$ .* From now on we write  $(M, \pi)$  for a Poisson manifold and use the bracket and the bivector interchangeably.

**Example 2.1** (Symplectic manifolds). Every symplectic manifold  $(M, \omega)$  is a Poisson manifold. The bracket  $\{f, g\} = \omega(X_f, X_g)$  is antisymmetric and Leibniz by construction, and satisfies the Jacobi identity because  $d\omega = 0$ . Comparing  $X_H = (\omega^\flat)^{-1}(dH)$  with  $X_H = \pi^\sharp(dH)$  identifies its bivector as the inverse of the form,

$$\pi^\sharp = (\omega^\flat)^{-1},$$

so here  $\pi^\sharp$  is an isomorphism at every point. By Darboux’s theorem [2] every symplectic manifold looks, near any point, like  $\mathbb{R}^{2n}$  with coordinates  $(q^1, \dots, q^n, p_1, \dots, p_n)$  in which

$$\{q^i, p_j\} = \delta_j^i, \quad \{q^i, q^j\} = \{p_i, p_j\} = 0,$$

that is  $\pi = \sum_i \partial_{q^i} \wedge \partial_{p_i}$ . The Hamiltonian vector field of  $H$  reproduces Hamilton’s equations,

$$\dot{q}^i = \{q^i, H\} = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = \{p_i, H\} = -\frac{\partial H}{\partial q^i}.$$

*Remark 2.2* (Sign conventions). The signs of Poisson geometry are not standardised. We have fixed  $X_H = \pi^\sharp dH$  and  $\dot{f} = \{f, H\}$  along the flow (Section 2.2.1). And on a symplectic manifold  $\{f, g\} = \omega(X_f, X_g)$  as above. Other references reverse the bracket or the sign of  $X_H$ . Either reversal flips the Lie–Poisson equation and the sign in the Koszul bracket. A comparison with another text should begin by matching these conventions.

*Remark 2.3* (Rank). The *rank* of a Poisson manifold at  $x$  is the rank of the linear map  $\pi_x^\sharp: T_x^*M \rightarrow T_xM$ , or equivalently of the antisymmetric matrix  $(\pi^{ij}(x))$  of components of  $\pi$  in a chart. Being the rank of an antisymmetric map, it is always even. The symplectic Poisson structures are exactly those of maximal rank  $\dim M$  at every point, which forces  $\dim M$  to be even. The zero bracket has rank 0 everywhere. In general the rank lies between these extremes and may change from point to point.

## 2.2. Casimirs and the symplectic foliation

The degeneracy of a Poisson bivector has two consequences that a symplectic structure does not show. Both follow from the Jacobi identity. Some functions are conserved by *every* Hamiltonian. They come from the kernel of  $\pi^\sharp$  and are called the Casimir functions. The manifold decomposes into symplectic submanifolds, the *leaves*, on which the trajectories stay. They come from the image of  $\pi^\sharp$ . We treat both only as far as the later chapters need.

### 2.2.1. Hamiltonian flows and Poisson maps

On a Poisson manifold every  $H \in C^\infty(M)$  generates the flow of  $X_H = \pi^\sharp(dH)$ , along which  $\dot{f} = \{f, H\}$ . Antisymmetry gives conservation of energy,  $\dot{H} = \{H, H\} = 0$ . More generally  $f$  is a constant of the motion for  $H$  when  $\{f, H\} = 0$ . The Jacobi identity adds two facts:

**Proposition 2.1.** *On a Poisson manifold, for all  $f, g, H \in C^\infty(M)$ ,*

$$(i) \quad X_H\{f, g\} = \{X_H f, g\} + \{f, X_H g\};$$

$$(ii) \quad [X_f, X_g] = -X_{\{f, g\}}.$$

Both are restatements of the Jacobi identity.

*Proof.* Write the Jacobi identity in the form  $\{a, \{b, c\}\} = \{\{a, b\}, c\} + \{b, \{a, c\}\}$ . For (i), take  $a = H$ ,  $b = f$ ,  $c = g$  and use  $X_H \phi = \{\phi, H\}$  together with antisymmetry:

$$X_H \{f, g\} = \{\{f, g\}, H\} = -\{H, \{f, g\}\} = -\{\{H, f\}, g\} - \{f, \{H, g\}\} = \{X_H f, g\} + \{f, X_H g\}.$$

For (ii), apply both sides to an arbitrary  $h$ . Since  $X_\phi \psi = \{\psi, \phi\}$ ,

$$[X_f, X_g]h = \{\{h, g\}, f\} - \{\{h, f\}, g\} = -\{h, \{f, g\}\} = -X_{\{f, g\}}h,$$

where the middle equality is the Jacobi identity.  $\square$

Part (i) says that  $X_H$  preserves the Poisson bracket. In tensor form this is  $\mathcal{L}_{X_H} \pi = 0$ . That is because the Schouten formula for the Lie derivative of a bivector [6] gives, for any vector field  $X$ ,

$$(\mathcal{L}_X \pi)(df, dg) = X\{f, g\} - \{Xf, g\} - \{f, Xg\},$$

which for  $X = X_H$  vanishes by Proposition 2.1(i). A tensor whose Lie derivative along  $X_H$  vanishes is invariant under the flow, so the flow  $\phi_t$  of  $X_H$  satisfies  $(\phi_t)_* \pi = \pi$  wherever it is defined. Maps that preserve the bracket have a name.

**Definition 2.2** (Poisson map). A smooth map  $\Phi: (M, \pi_M) \rightarrow (N, \pi_N)$  between Poisson manifolds is a *Poisson map* if

$$\{\varphi \circ \Phi, \psi \circ \Phi\}_M = \{\varphi, \psi\}_N \circ \Phi \quad \text{for all } \varphi, \psi \in C^\infty(N).$$

For a diffeomorphism, preservation of  $\pi$  is exactly this condition. Every Hamiltonian flow is therefore a Poisson diffeomorphism between its open domains of definition. Poisson maps are the natural morphisms between Poisson manifolds.

### 2.2.2. Casimir functions

The kernel of  $\pi^\sharp$  carries the functions that are constant for any dynamics.

**Definition 2.3** (Casimir function). A *Casimir function* of  $(M, \pi)$  is a function  $C \in C^\infty(M)$  with  $\{C, f\} = 0$  for all  $f \in C^\infty(M)$ .

Since  $\{C, f\} = -\langle df, \pi^\sharp dC \rangle$ , a function  $C$  is a Casimir exactly when  $\pi^\sharp dC = 0$ . That is  $X_C = 0$ , or equivalently  $dC \in \ker \pi^\sharp$  at every point. The dynamical meaning is that for *every* Hamiltonian  $H$ ,

$$\dot{C} = \{C, H\} = 0.$$

A Casimir is therefore conserved by every Hamiltonian system on  $(M, \pi)$  at once. On a symplectic manifold  $\pi^\sharp$  is injective, so Casimir functions are locally constant, and they are constant when the manifold is connected. More generally, a Casimir with nonzero differential detects degeneracy at that point. On a regular Poisson chart of dimension  $n$  and rank  $2r$  there are  $n - 2r$  independent local Casimirs, whereas degeneracy alone does not guarantee nonconstant global Casimirs or an analogous statement at singular points.

### 2.2.3. The symplectic foliation

The image of  $\pi^\sharp$  carries the directions available to the dynamics. At each point  $x$ , as  $f$  ranges over  $C^\infty(M)$ , the vectors  $X_f(x) = \pi^\sharp(df_x)$  fill the subspace

$$\mathcal{D}_x = \text{im } \pi_x^\sharp \subseteq T_x M.$$

The assignment  $x \mapsto \mathcal{D}_x$  is the *characteristic distribution*. Its dimension is the rank of  $\pi$ , which may change from point to point. Consequently the Frobenius theorem does not apply.

The characteristic distribution  $\mathcal{D} = \text{im } \pi^\sharp$  is generated by the family of Hamiltonian vector fields  $F = \{X_f : f \in C^\infty(M)\}$ : at each  $x$ ,  $\mathcal{D}_x = \{X_f(x) : f \in C^\infty(M)\}$ , since every covector is a differential. This family is closed under the Lie bracket, by Proposition 2.1(ii), and  $\mathcal{D}$  is invariant under the flows of its members, by Proposition 2.1(i). For a family of vector fields with these properties the generated distribution has non-constant rank in general, so the Frobenius theorem does not apply; integrability is instead guaranteed by the Stefan–Sussmann theorem [6, 15].

**Theorem 2.1** (Symplectic foliation). *Let  $(M, \pi)$  be a Poisson manifold.*

(i) *The characteristic distribution integrates to a foliation of  $M$  by immersed submanifolds, the symplectic leaves; the leaf through  $x$  has tangent space  $\mathcal{D}_x$ , and two points lie on the same leaf exactly when a chain of Hamiltonian flows joins them.*

(ii) *Each leaf  $L$  carries a unique symplectic form, given by*

$$\omega_L(\pi^\sharp \alpha, \pi^\sharp \beta) = \pi(\alpha, \beta),$$

*for which the inclusion  $L \hookrightarrow M$  is a Poisson map.*

We take part (i) from [6, 15], as we do the closedness of  $\omega_L$  in part (ii), which is where the Jacobi identity of  $\pi$  in the Schouten bracket form is used. That  $\omega_L$  is well defined and non-degenerate is pure linear algebra:  $\pi^\sharp \alpha = \pi^\sharp \alpha'$  gives  $\pi(\alpha - \alpha', \beta) = -\langle \beta, \pi^\sharp(\alpha - \alpha') \rangle = 0$ , non-degenerate because  $\omega_L(\pi^\sharp \alpha, \cdot) = 0$  forces  $\pi(\alpha, \cdot) = 0$  and hence  $\pi^\sharp \alpha = 0$ . Since  $T_x L = \mathcal{D}_x = \text{im } \pi_x^\sharp$  by part (i), the dimension of  $L$  equals the rank of  $\pi$  on  $L$ . It is even because  $\pi^\sharp$  is antisymmetric.

Since  $X_H$  takes values in  $\mathcal{D}$ , every trajectory of  $H$  stays on the leaf through its initial point. On that leaf it is the symplectic flow of the restriction  $H|_L$ . A Casimir  $C$  is constant on each leaf: for  $v = \pi^\sharp \alpha \in \mathcal{D}_x$ ,  $dC(v) = \langle dC, \pi^\sharp \alpha \rangle = -\langle \alpha, \pi^\sharp dC \rangle = 0$ , since  $\pi^\sharp dC = 0$ . A connected symplectic manifold is the case of a single leaf.

**Example 2.2** (see [6]). On  $\mathbb{R}^3$  with coordinates  $(x, y, z)$  take  $\pi = \partial_x \wedge \partial_y$ , so that

$$\{x, y\} = 1, \quad \{x, z\} = \{y, z\} = 0.$$

The rank is 2 everywhere. The characteristic distribution is spanned by  $\partial_x$  and  $\partial_y$ , and  $z$  is a Casimir, since  $dz$  spans  $\ker \pi^\sharp$  at every point. No Hamiltonian moves  $z$ , so every trajectory stays in a plane  $z = \text{const}$ , on which  $\pi$  restricts to the standard symplectic structure of  $\mathbb{R}^2$ . The symplectic leaves are these planes, which are the level sets of the single Casimir  $z$ .

## 2.3. The Lie–Poisson structure and the rigid body

Sections 2.1–2.2 were general theory. We now study one specific Poisson manifold, the dual  $\mathfrak{g}^*$  of a Lie algebra. It is the phase space of mechanical systems reduced by symmetry; the free rigid body is the main case considered here. Its cotangent algebroid, defined in Section 2.4, is integrated by the groupoid  $T^*G \rightrightarrows \mathfrak{g}^*$  used in the latter chapters.

### 2.3.1. The Lie–Poisson bracket

Let  $\mathfrak{g}$  be a finite-dimensional Lie algebra with bracket  $[\cdot, \cdot]$ , and  $\mathfrak{g}^*$  its dual. Because  $\mathfrak{g}^*$  is a vector space, its tangent space at any point is canonically again  $\mathfrak{g}^*$ , and its cotangent space is the double dual,

$$T_\mu \mathfrak{g}^* \cong \mathfrak{g}^*, \quad T_\mu^* \mathfrak{g}^* \cong (\mathfrak{g}^*)^* \cong \mathfrak{g}.$$

It follows that the differential of a function  $f \in C^\infty(\mathfrak{g}^*)$  at a point  $\mu$  is an element of  $\mathfrak{g}$ ,

$$d_\mu f \in \mathfrak{g}, \quad \langle \nu, d_\mu f \rangle = \left. \frac{d}{dt} \right|_{t=0} f(\mu + t\nu) \quad (\nu \in \mathfrak{g}^*),$$

where  $\langle \cdot, \cdot \rangle$  is the pairing between  $\mathfrak{g}^*$  and  $\mathfrak{g}$ . For a linear function  $\ell_\xi(\mu) = \langle \mu, \xi \rangle$  with  $\xi \in \mathfrak{g}$ , the differential is constant,  $d_\mu \ell_\xi = \xi$ . This identification allows us to define the Poisson structure on  $\mathfrak{g}^*$  based on the Lie algebra structure of  $\mathfrak{g}$ .

**Definition 2.4** (Lie–Poisson bracket). The *Lie–Poisson bracket* on  $\mathfrak{g}^*$  is

$$\{f, g\}(\mu) = \langle \mu, [d_\mu f, d_\mu g] \rangle, \quad f, g \in C^\infty(\mathfrak{g}^*).$$

The bracket is bilinear and antisymmetric. It is a derivation in each argument, since  $d_\mu(fg) = f(\mu) d_\mu g + g(\mu) d_\mu f$  and the pairing is linear. On linear functions it returns the Lie bracket,

$$\{\ell_\xi, \ell_\eta\} = \ell_{[\xi, \eta]},$$

because  $d\ell_\xi = \xi$  and  $d\ell_\eta = \eta$  are constant. Fixing a basis  $\{e_i\}$  of  $\mathfrak{g}$  with  $[e_i, e_j] = c_{ij}^k e_k$  and writing  $\mu_i = \ell_{e_i}$  for the resulting linear coordinates on  $\mathfrak{g}^*$ ,

$$\{\mu_i, \mu_j\} = c_{ij}^k \mu_k, \quad \text{so} \quad \{f, g\}(\mu) = c_{ij}^k \mu_k \frac{\partial f}{\partial \mu_i} \frac{\partial g}{\partial \mu_j}.$$

The bracket is thus linear in  $\mu$  and vanishes at the origin.

It remains to verify the Jacobi identity. The Jacobiator

$$J(f, g, h) = \{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\}$$

is a derivation in each argument. By Remark 2.1 it vanishes identically as soon as it vanishes on the coordinates  $\mu_i$ . There,

$$J(\mu_i, \mu_j, \mu_k) = (c_{jk}^m c_{im}^\ell + c_{ki}^m c_{jm}^\ell + c_{ij}^m c_{km}^\ell) \mu_\ell.$$

The coefficient is exactly the Jacobi identity for the structure constants of  $\mathfrak{g}$ , hence zero. Therefore, the Lie–Poisson bracket satisfies the Jacobi identity, and  $(\mathfrak{g}^*, \{\cdot, \cdot\})$  is a Poisson manifold.

### 2.3.2. The equations of motion

On  $\mathfrak{g}^*$  the Hamiltonian equations  $\dot{f} = \{f, H\}$  take a form written entirely inside the Lie algebra. For  $\xi \in \mathfrak{g}$  recall the *adjoint* operator  $\text{ad}_\xi \eta = [\xi, \eta]$  and define the *coadjoint* operator  $\text{ad}_\xi^*: \mathfrak{g}^* \rightarrow \mathfrak{g}^*$  by

$$\langle \text{ad}_\xi^* \mu, \eta \rangle = \langle \mu, [\xi, \eta] \rangle, \quad \eta \in \mathfrak{g}.$$

With this convention  $\text{ad}_\xi^*$  is the linear transpose of  $\text{ad}_\xi$ , whereas the group-level coadjoint action is defined using  $\text{Ad}_{g^{-1}}$ . Consequently, the infinitesimal generator of the coadjoint action is  $-\text{ad}_\xi^*$ . Testing  $\dot{\mu}$  against a constant  $\xi = d\ell_\xi$  and using antisymmetry of the Lie bracket,

$$\langle \dot{\mu}, \xi \rangle = \left. \frac{d}{dt} \right|_{t=0} \ell_\xi(\mu(t)) = \{\ell_\xi, H\}(\mu) = \langle \mu, [\xi, d_\mu H] \rangle = -\langle \mu, [d_\mu H, \xi] \rangle = -\langle \text{ad}_{d_\mu H}^* \mu, \xi \rangle.$$

As  $\xi$  is arbitrary, this is the *Lie–Poisson equation*

$$\dot{\mu} = -\text{ad}_{d_\mu H}^* \mu. \quad (2.3.1)$$

The entire dynamics is defined using the coadjoint operator and the single element  $d_\mu H \in \mathfrak{g}$ .

### 2.3.3. Coadjoint orbits and the symplectic leaves

The group  $G$  acts on its Lie algebra by the *adjoint representation*  $\text{Ad}_g = T_e C_g$ , the differential at the identity of the conjugation  $C_g(k) = gkg^{-1}$ . From  $C_{gh} = C_g \circ C_h$  and  $C_g(e) = e$  one gets  $\text{Ad}_{gh} = \text{Ad}_g \text{Ad}_h$ . Dually  $G$  acts on  $\mathfrak{g}^*$  by the *coadjoint action*

$$\langle \text{Ad}_g^* \mu, \xi \rangle = \langle \mu, \text{Ad}_{g^{-1}} \xi \rangle.$$

Its orbits  $\mathcal{O}_\mu = \{\text{Ad}_g^* \mu : g \in G\}$  are called the *coadjoint orbits*. Differentiating along  $\exp(t\xi)$  gives the vector fields generating the action,

$$\left. \frac{d}{dt} \right|_{t=0} \text{Ad}_{\exp(t\xi)}^* \mu = -\text{ad}_\xi^* \mu.$$

The tangent space to the orbit through  $\mu$  is  $T_\mu \mathcal{O}_\mu = \{\text{ad}_\xi^* \mu : \xi \in \mathfrak{g}\}$ .

This is exactly the characteristic distribution. The cotangent space  $T_\mu^* \mathfrak{g}^*$  is  $\mathfrak{g}$ , spanned by the differentials  $d\ell_\xi = \xi$ . Therefore  $\text{im } \pi_\mu^\sharp$  is spanned by the Hamiltonian vector fields of the linear functions. By (2.3.1) with  $H = \ell_\xi$  the field  $X_{\ell_\xi}$  takes the value  $-\text{ad}_\xi^* \mu$  at  $\mu$ . Hence,

$$\text{im } \pi_\mu^\sharp = \{\text{ad}_\xi^* \mu : \xi \in \mathfrak{g}\} = T_\mu \mathcal{O}_\mu.$$

The distribution matches the tangent spaces of the orbits. That the leaves are the connected components of the coadjoint orbits is the theorem of Kirillov, Kostant and Souriau. We take it from [6, 11].

**Theorem 2.2** (Kirillov–Kostant–Souriau). *The symplectic leaves of  $\mathfrak{g}^*$  are the connected components of the coadjoint orbits. On the orbit through  $\mu$  the leaf symplectic form is*

$$\omega_\mu(\text{ad}_\xi^* \mu, \text{ad}_\eta^* \mu) = \langle \mu, [\xi, \eta] \rangle, \quad \xi, \eta \in \mathfrak{g}.$$

The formula is the one produced by Theorem 2.1(ii): evaluating  $\omega_L(\pi^\sharp d\ell_\xi, \pi^\sharp d\ell_\eta) = \pi(d\ell_\xi, d\ell_\eta)$  gives  $\{\ell_\xi, \ell_\eta\}(\mu) = \langle \mu, [\xi, \eta] \rangle$ , and the two signs from  $\pi^\sharp d\ell_\xi = -\text{ad}_\xi^* \mu$  cancel.

A Casimir is constant on each leaf, hence on each connected component of a coadjoint orbit. This shows that it is invariant under the identity component  $G_0$ . When  $G$  is connected, as for  $G = \text{SO}(3)$ , the components are the full orbits and the Casimirs are exactly the  $\text{Ad}^*$ -invariant functions.

### 2.3.4. The free rigid body

The rotating rigid body is the case  $G = \text{SO}(3)$ . Its configuration is the rotation  $R \in \text{SO}(3)$  that carries the body frame to the spatial frame. The spatial frame is fixed in the laboratory. The body frame is attached to the body and turns with it. The choice of frame gives two descriptions of the same motion. In the spatial frame the angular momentum is conserved. In the body frame it is  $\mu = I\omega$  with angular velocity  $\omega = (\omega_1, \omega_2, \omega_3)$  and constant inertia tensor  $I$ . The body frame is convenient because  $I$  is constant there. The phase space is  $T^*\text{SO}(3)$  of dimension six. The energy is kinetic only and depends on  $\omega$  and not on the orientation. It is thus invariant under the rotational symmetry. Reduction by that symmetry leaves three coordinates, the body angular momentum  $\mu \in \mathfrak{so}(3)^*$ . The reduction is carried out in Chapter 4. Here we first read the dynamics on  $\mathfrak{so}(3)^*$  directly.

We identify both  $\mathfrak{so}(3)$  and  $\mathfrak{so}(3)^*$  with  $\mathbb{R}^3$ . This way the bracket, the adjoint and coadjoint operators and the Lie–Poisson bracket all become vector operations. The first identification is the hat map [11]

$$\hat{\cdot} : \mathbb{R}^3 \rightarrow \mathfrak{so}(3), \quad \hat{\xi} \eta = \xi \times \eta.$$

It sends a vector  $\xi$  to the matrix that acts as the cross product with  $\xi$ . It is a linear isomorphism which respects the brackets. For any  $\zeta$

$$[\hat{\xi}, \hat{\eta}] \zeta = \xi \times (\eta \times \zeta) - \eta \times (\xi \times \zeta) = (\xi \times \eta) \times \zeta = \widehat{\xi \times \eta} \zeta,$$

where the middle step is the Jacobi identity of the cross product. Consequently  $[\hat{\xi}, \hat{\eta}] = \widehat{\xi \times \eta}$  and the hat map is a Lie algebra isomorphism from  $(\mathbb{R}^3, \times)$  onto  $\mathfrak{so}(3)$ . The adjoint operator is then the cross product,

$$\mathrm{ad}_\xi \eta = [\xi, \eta] = \xi \times \eta,$$

by the bracket identity just proved.

Rotations preserve the cross product,  $R(a \times b) = (Ra) \times (Rb)$  for  $R \in \mathrm{SO}(3)$ . Acting on a vector  $v$  we get  $R\hat{\xi}R^{-1}v = R(\xi \times R^{-1}v) = (R\xi) \times v$ . It follows that

$$\mathrm{Ad}_R \hat{\xi} = \widehat{R\xi}.$$

Conjugation on  $\mathfrak{so}(3)$  is rotation of the underlying vector.

The second identification uses the dot product. It is a nondegenerate pairing, so every functional on  $\mathfrak{so}(3) \cong \mathbb{R}^3$  is  $\eta \mapsto m \cdot \eta$  for a unique  $m \in \mathbb{R}^3$ . This identifies  $\mathfrak{so}(3)^*$  with  $\mathbb{R}^3$ ,

$$\langle \mu, \hat{\eta} \rangle = m \cdot \eta, \quad \mu \leftrightarrow m.$$

The dot product is also  $\mathrm{SO}(3)$ -invariant since  $R$  is orthogonal. This is what turns the coadjoint action into a rotation:

$$\langle \mathrm{Ad}_R^* \mu, \hat{\xi} \rangle = \langle \mu, \mathrm{Ad}_{R^{-1}} \hat{\xi} \rangle = \langle \mu, \widehat{R^{-1}\xi} \rangle = m \cdot (R^{-1}\xi) = (Rm) \cdot \xi,$$

using  $R^{-1} = R^\top$ . Hence,  $\mathrm{Ad}_R^* \mu = R\mu$ . The coadjoint action is the standard rotation action and its orbits are the spheres  $|\mu| = \text{const}$  and the origin.

The coadjoint operator follows. By its definition

$$\langle \mathrm{ad}_\xi^* \mu, \eta \rangle = \langle \mu, [\xi, \eta] \rangle = \mu \cdot (\xi \times \eta) = (\mu \times \xi) \cdot \eta,$$

where the last step is the scalar triple product identity. This holds for every  $\eta$  so

$$\mathrm{ad}_\xi^* \mu = \mu \times \xi.$$

It remains to find the bracket. A function  $f$  on  $\mathfrak{so}(3)^*$  has a differential  $d_\mu f \in \mathfrak{g}$  by Section 2.3.1. The pairing is now the dot product so

$$\nu \cdot d_\mu f = \left. \frac{d}{dt} \right|_{t=0} f(\mu + t\nu) = \nu \cdot \nabla f,$$

hence  $d_\mu f = \nabla f$  is the ordinary gradient. Placing this into Definition 2.4 gives the Lie–Poisson bracket in vector form

$$\{f, g\}(\mu) = \langle \mu, [d_\mu f, d_\mu g] \rangle = \mu \cdot (\nabla f \times \nabla g).$$

Thus, the bracket on  $\mathfrak{so}(3)^*$  is precisely Definition 2.4 expressed in vector notation.

*Remark 2.4* (Body frame and the sign). The bracket of the rigid body is not Definition 2.4 itself. The body lives on  $T^*\mathrm{SO}(3)$  with the canonical bracket. Its momentum  $\mu$  is the canonical momentum read in the body frame, carried to the identity by left translation. Reducing the canonical bracket in this variable gives the *minus* Lie–Poisson bracket

$$\{f, g\}(\mu) = -\langle \mu, [d_\mu f, d_\mu g] \rangle.$$

The spatial frame would give the plus bracket instead. Both share leaves, Casimirs and orbits. But the equation of motion keeps the sign  $\dot{\mu} = +\mathrm{ad}_{d_\mu H}^* \mu$ . We adopt the minus bracket and derive the reduction in Chapter 4. Compare [11].

On  $\mathfrak{so}(3)^* \cong \mathbb{R}^3$  the minus bracket is

$$\{f, g\}(\mu) = -\mu \cdot (\nabla f \times \nabla g).$$

The function  $C(\mu) = \frac{1}{2}|\mu|^2$  is a Casimir. Indeed  $\nabla C = \mu$  and  $\{C, f\} = -\mu \cdot (\mu \times \nabla f) = 0$ . Its level sets are the spheres, as Theorem 2.2 requires. A body with principal moments of inertia  $I_1, I_2, I_3$  has energy

$$H(\mu) = \frac{1}{2} \mu \cdot I^{-1} \mu, \quad I = \mathrm{diag}(I_1, I_2, I_3),$$

and  $\nabla H = I^{-1}\mu = \omega$  is the angular velocity. The equation of motion  $\dot{\mu} = \text{ad}_{\nabla H}^* \mu = \mu \times \nabla H$  becomes Euler's equation

$$\dot{\mu} = \mu \times \omega. \quad (2.3.2)$$

With  $\mu = I\omega$  the first component is  $I_1\dot{\omega}_1 = (I_2 - I_3)\omega_2\omega_3$ , and the cyclic rotation  $1 \rightarrow 2 \rightarrow 3$  gives the other two. This is the classical Euler equation and it confirms the sign of the remark.

## 2.4. The cotangent Lie algebroid

Chapter 1 built the Lie algebroid of a Lie groupoid by differentiation at the units. Chapter 3 studies symplectic groupoids that integrate Poisson manifolds. In preparation, this section builds an algebroid on the cotangent bundle  $T^*M$  of a Poisson manifold  $(M, \pi)$ .

### 2.4.1. Anchor and bracket

We look for a Lie algebroid structure on  $A = T^*M$ . By Definition 1.3 it is an anchor  $\rho: T^*M \rightarrow TM$  and a Lie bracket on  $\Omega^1(M)$  with the Leibniz rule. Two requirements fix both.

The anchor is fixed by the dynamics. We demand that the algebroid generates the Hamiltonian flows, so we require  $\rho(dH) = X_H = \pi^\sharp(dH)$  for every  $H$ . At a point  $x$  every covector is  $d_x H$  for some  $H$ , and  $\rho_x$  is linear, so this forces  $\rho = \pi^\sharp$ .

The bracket is fixed on exact forms. On the differentials it must return the Poisson bracket of the functions. If  $(T^*M, \pi^\sharp, [\cdot, \cdot])$  is an algebroid then its anchor preserves brackets (Proposition 1.1). Therefore

$$\pi^\sharp[df, dg] = [\pi^\sharp df, \pi^\sharp dg] = [X_f, X_g] = -X_{\{f, g\}} = \pi^\sharp(-d\{f, g\}),$$

by Proposition 2.1(ii). This motivates the canonical choice

$$[df, dg]_\pi = -d\{f, g\}. \quad (2.4.1)$$

The Leibniz rule then carries the bracket from exact forms to all of  $\Omega^1(M)$ , because locally every one-form is a combination  $\sum_i f_i dg_i$ . Doing this gives a closed formula, which we take as the definition.

**Definition 2.5** (Koszul bracket). Let  $(M, \pi)$  be a Poisson manifold. The *Koszul bracket* on  $\Omega^1(M)$  is

$$[\alpha, \beta]_\pi = \mathcal{L}_{\pi^\sharp \alpha} \beta - \mathcal{L}_{\pi^\sharp \beta} \alpha + d(\pi(\alpha, \beta)).$$

On exact forms it reduces to (2.4.1). Using  $\pi^\sharp df = X_f$  with  $X_f(h) = \{h, f\}$  and  $\mathcal{L}_X dh = d(Xh)$ ,

$$[df, dg]_\pi = d\{g, f\} - d\{f, g\} + d\{f, g\} = -d\{f, g\}.$$

For any bivector  $\pi$  the formula is antisymmetric. It also obeys the Leibniz rule with anchor  $\pi^\sharp$ . With  $\mathcal{L}_{hX}\alpha = h\mathcal{L}_X\alpha + \alpha(X)dh$  and  $\pi^\sharp(h\beta) = h\pi^\sharp\beta$  we get

$$[\alpha, h\beta]_\pi = h[\alpha, \beta]_\pi + (\pi^\sharp \alpha \cdot h)\beta + (\pi(\alpha, \beta) - \alpha(\pi^\sharp \beta))dh.$$

The last term is zero because  $\alpha(\pi^\sharp \beta) = \pi(\alpha, \beta)$  is the definition of  $\pi^\sharp$ . This shows that antisymmetry and the Leibniz rule hold for every bivector. Only the Jacobi identity can fail. In Section 2.1 the single condition on  $\pi$  was the Jacobi identity of the bracket, that is  $[\pi, \pi]_{\text{SN}} = 0$ . The Jacobi identity of the Koszul bracket is the same condition.

**Theorem 2.3.** Let  $(M, \pi)$  be a Poisson manifold. Then  $T^*M$ , with anchor  $\pi^\sharp$  and the Koszul bracket, is a Lie algebroid. It is the unique Lie algebroid structure on  $T^*M$  with anchor  $\pi^\sharp$  that restricts to (2.4.1) on exact forms. We call it the cotangent Lie algebroid of  $(M, \pi)$ .

*Proof.* Antisymmetry and the Leibniz rule hold for any bivector. We showed this above. Only the Jacobi identity needs  $\pi$  to be Poisson. We prove it in three steps.

First, we check that the anchor respects the bracket. We claim that

$$\pi^\sharp[\alpha, \beta]_\pi = [\pi^\sharp\alpha, \pi^\sharp\beta] \quad (2.4.2)$$

for all one-forms  $\alpha$  and  $\beta$ . Both sides are antisymmetric. Under  $\beta \mapsto h\beta$  each side gains the term  $(\pi^\sharp\alpha \cdot h)\pi^\sharp\beta$  by the Leibniz rule of the corresponding bracket. Consequently, the difference of the two sides is  $C^\infty(M)$ -bilinear. Then it is enough to check (2.4.2) on exact forms. There it becomes  $[X_f, X_g] = -X_{\{f, g\}}$ , which is Proposition 2.1(ii).

Second, we check that the Jacobiator

$$J(\alpha, \beta, \gamma) = [[\alpha, \beta]_\pi, \gamma]_\pi + [[\beta, \gamma]_\pi, \alpha]_\pi + [[\gamma, \alpha]_\pi, \beta]_\pi$$

is a tensor. It is antisymmetric, so it is enough to test the last argument. Replace  $\gamma$  by  $h\gamma$  and expand by the Leibniz rule. The terms carrying no derivative of  $h$  add up to  $hJ(\alpha, \beta, \gamma)$ . The terms carrying one derivative of  $h$  cancel in pairs by antisymmetry. The terms carrying two derivatives of  $h$  and a factor  $\gamma$  add up to

$$(\pi^\sharp[\alpha, \beta]_\pi - [\pi^\sharp\alpha, \pi^\sharp\beta])(h)\gamma.$$

This vanishes because the anchor respects the bracket. It follows that  $J(\alpha, \beta, h\gamma) = hJ(\alpha, \beta, \gamma)$ . Similar computation appears in Proposition 1.1. Here we make it for the Koszul bracket.

A tensor is fixed by its values on exact forms. They span  $\Omega^1(M)$  over  $C^\infty(M)$ . On exact forms using (2.4.1) twice we get

$$J(df, dg, dh) = d(\{\{f, g\}, h\} + \{\{g, h\}, f\} + \{\{h, f\}, g\}) = 0$$

by the Jacobi identity for  $\{\cdot, \cdot\}$ . Hence  $J \equiv 0$ . The Koszul bracket is a Lie bracket and  $(T^*M, \pi^\sharp, [\cdot, \cdot]_\pi)$  is a Lie algebroid.

For uniqueness, take two Lie algebroid brackets on  $T^*M$  with anchor  $\pi^\sharp$  that both restrict to (2.4.1) on exact forms. Both are antisymmetric, and under  $\beta \mapsto h\beta$  each gains the same term  $(\pi^\sharp\alpha \cdot h)\beta$  by the Leibniz rule. Therefore their difference is  $C^\infty(M)$ -bilinear, as in the first step. It vanishes on exact forms, which span, so the two brackets agree.  $\square$

## 2.4.2. Examples

**Example 2.3** (Symplectic manifolds). Let  $(M, \omega)$  be symplectic, so  $\pi^\sharp = (\omega^\flat)^{-1}$  is an isomorphism (Example 2.1). It is an isomorphism of Lie algebroids between  $T^*M$  and the tangent algebroid  $(TM, [\cdot, \cdot], \text{id})$ , which is the algebroid of the pair groupoid (Examples 1.2 and 1.5). The anchors agree because  $\text{id} \circ \pi^\sharp = \pi^\sharp$ . On exact forms the brackets agree, since

$$\pi^\sharp[df, dg]_\pi = \pi^\sharp(-d\{f, g\}) = -X_{\{f, g\}} = [X_f, X_g] = [\pi^\sharp df, \pi^\sharp dg],$$

and the two minus signs cancel. This shows that the cotangent algebroid of a symplectic manifold is canonically its tangent bundle. We return to this result in Chapter 3.

**Example 2.4** (The dual of a Lie algebra). Let  $\mathfrak{g}^*$  carry the Lie–Poisson structure of Definition 2.4. It is a vector space, so  $T_\mu^*\mathfrak{g}^* \cong \mathfrak{g}$  at every point. The cotangent bundle is trivial,

$$T^*\mathfrak{g}^* = \mathfrak{g}^* \times \mathfrak{g}.$$

A one-form is therefore a  $\mathfrak{g}$ -valued function on  $\mathfrak{g}^*$ . The constant ones are the elements of  $\mathfrak{g}$  (Section 2.3.1). Each of them is exact. That is because  $d_\mu \ell_\xi = \xi$  for every  $\mu$ , so the constant section  $\xi$  is the form  $d\ell_\xi$ . These sections span the fibre at every point.

The anchor of the cotangent algebroid is  $\pi^\sharp$  (Theorem 2.3). By definition  $\pi^\sharp dH = X_H$ , so the anchor sends  $d\ell_\xi$  to  $X_{\ell_\xi}$ . Its value follows from the Lie–Poisson equation (2.3.1) with  $H = \ell_\xi$ ,

$$\pi^\sharp d\ell_\xi|_\mu = X_{\ell_\xi}(\mu) = -\text{ad}_\xi^* \mu.$$

By Section 2.3.3 this is the value at  $\mu$  of the vector field generating the coadjoint action, written  $a(\xi)_\mu$  in the notation of Example 1.6.

The bracket of two constant sections follows from (2.4.1) and  $\{\ell_\xi, \ell_\eta\} = \ell_{[\xi, \eta]}$ ,

$$[d\ell_\xi, d\ell_\eta]_\pi = -d\{\ell_\xi, \ell_\eta\} = -d\ell_{[\xi, \eta]}.$$

Consequently, on constant sections the bracket is the bracket of  $\mathfrak{g}$  with a minus sign. The sign is forced by the anchor. The anchor must carry the bracket of forms to the bracket of fields (Proposition 1.1). By Proposition 2.1(ii)  $[X_f, X_g] = -X_{\{f, g\}}$ , so the bracket of exact forms must be  $-d\{f, g\}$ .

The action algebroid  $\mathfrak{g} \ltimes \mathfrak{g}^*$  of the coadjoint action carries the opposite signs. Example 1.6 computed its anchor on constant sections as  $-a$  and its bracket on constant sections as  $[\xi, \eta]$ . It follows that the two algebroids over  $\mathfrak{g}^*$  differ by a sign in the anchor and by a sign in the bracket. The differences come from two conventions that were fixed independently of each other. First is the convention  $A(\mathcal{G}) = \ker(Tt)$  of Section 1.3.1. The second is the convention  $X_H = \{\cdot, H\}$  of Remark 2.2, through Proposition 2.1(ii). One map removes both differences. Put

$$\Phi: T^*\mathfrak{g}^* \rightarrow \mathfrak{g} \ltimes \mathfrak{g}^*, \quad \Phi(\mu, \xi) = (\mu, -\xi).$$

Then, the anchors agree,

$$\rho(\Phi(\mu, \xi)) = -a(-\xi)_\mu = a(\xi)_\mu = \pi^\# d\ell_\xi|_\mu,$$

and so do the brackets of constant sections,

$$[\Phi\xi, \Phi\eta] = [-\xi, -\eta] = [\xi, \eta] = \Phi(-[\xi, \eta]) = \Phi([d\ell_\xi, d\ell_\eta]_\pi).$$

Agreement extends from the constant sections to all sections. Write a section in a basis of  $\mathfrak{g}$ , with coefficients in  $C^\infty(\mathfrak{g}^*)$ . Both brackets are antisymmetric and expand through the Leibniz rule of Definition 1.3. In the expansions the coefficients multiply brackets of constant sections, which agree. The derivative terms contain only the anchors, which agree. Hence the two expansions are equal, and  $\Phi$  is an isomorphism of Lie algebroids. The cotangent algebroid of  $\mathfrak{g}^*$  is the action algebroid of the coadjoint action. We return to this result in Chapter 3.

The minus sign in  $\Phi$  records the two conventions together. With  $\ker(Ts)$  and right-invariant fields in Section 1.3.1 the action algebroid would have anchor  $a$  and bracket  $-[\xi, \eta]$  on constant sections, and  $\Phi$  would be the identity. With  $X_H = \{H, \cdot\}$  in Remark 2.2 the cotangent algebroid would have anchor  $-a$  and bracket  $[\xi, \eta]$ , and again  $\Phi$  would be the identity.

## Chapter 3

# Symplectic Groupoids

### 3.1. The definition and its Lagrangian form

**Definition 3.1** (Symplectic groupoid). A *symplectic groupoid* is a Lie groupoid  $\mathcal{G} \rightrightarrows M$  together with a symplectic form  $\Omega \in \Omega^2(\mathcal{G})$  that is *multiplicative*:

$$m^*\Omega = \text{pr}_1^*\Omega + \text{pr}_2^*\Omega \quad \text{on } \mathcal{G}^{(2)}, \quad (3.1.1)$$

where  $m: \mathcal{G}^{(2)} \rightarrow \mathcal{G}$  is the multiplication and  $\text{pr}_1, \text{pr}_2: \mathcal{G}^{(2)} \rightarrow \mathcal{G}$  are the two projections.

In Section 3.2 we show that  $\dim \mathcal{G} = 2 \dim M$ , that the units are Lagrangian, and derive other key structure consequences of this relation. On tangent vectors multiplicativity reads

$$\Omega_{gh}(Tm(X, X'), Tm(Y, Y')) = \Omega_g(X, Y) + \Omega_h(X', Y') \quad (3.1.2)$$

for all composable tangent pairs  $(X, X'), (Y, Y') \in T_{(g,h)}\mathcal{G}^{(2)}$ . A pair  $(X, X') \in T_g\mathcal{G} \times T_h\mathcal{G}$  is composable when  $Ts(X) = Tt(X')$ . Most symplectic forms on a Lie groupoid are not multiplicative.

**Proposition 3.1** (Lagrangian form of multiplicativity). *Let  $\mathcal{G} \rightrightarrows M$  be a Lie groupoid carrying a symplectic form  $\Omega$ , and let  $\overline{\mathcal{G}}$  denote  $\mathcal{G}$  with the symplectic form  $-\Omega$ . Then  $\Omega$  is multiplicative if and only if the graph of multiplication*

$$\text{Gr}(m) = \{(g, h, gh) : (g, h) \in \mathcal{G}^{(2)}\}$$

*is an isotropic submanifold of  $\mathcal{G} \times \mathcal{G} \times \overline{\mathcal{G}}$ .*

*Proof.* A tangent vector to  $\text{Gr}(m)$  at  $(g, h, gh)$  has the form  $(X, X', Tm(X, X'))$  for a composable pair  $(X, X')$ . The ambient symplectic form is  $\text{pr}_1^*\Omega + \text{pr}_2^*\Omega - \text{pr}_3^*\Omega$ , so two such tangent vectors pair to

$$\Omega_g(X, Y) + \Omega_h(X', Y') - \Omega_{gh}(Tm(X, X'), Tm(Y, Y')).$$

This vanishes for all composable pairs precisely when (3.1.2) holds. That is, precisely when  $\Omega$  is multiplicative.  $\square$

The proposition states only that  $\text{Gr}(m)$  is isotropic. Calling it Lagrangian requires it to have half the dimension of  $\mathcal{G} \times \mathcal{G} \times \overline{\mathcal{G}}$ . That count needs  $\dim \mathcal{G} = 2 \dim M$ , which is itself a consequence of multiplicativity. It is proved in Section 3.2 and the statement is completed there in Remark 3.1.

The following two examples run through the chapter. The first is the pair groupoid of a symplectic manifold. The second is the cotangent groupoid  $T^*G \rightrightarrows \mathfrak{g}^*$ . Chapter 4 reads the second example as Lie–Poisson reduction.

**Example 3.1** (The pair groupoid of a symplectic manifold). Let  $(M, \omega)$  be a symplectic manifold and take the pair groupoid  $M \times M \rightrightarrows M$  of Example 1.2, with  $s(y, x) = x$ ,  $t(y, x) = y$  and  $(z, y)(y, x) = (z, x)$ . Endow it with the two-form

$$\Omega = \text{pr}_1^* \omega - \text{pr}_2^* \omega, \quad \text{pr}_1(y, x) = y, \quad \text{pr}_2(y, x) = x.$$

It is closed because  $\omega$  is. For non-degeneracy split the tangent space at  $(y, x)$ ,

$$T_{(y,x)}(M \times M) = T_y M \oplus T_x M,$$

and write  $\omega_y$  and  $\omega_x$  for the values of  $\omega$  at the two points. These are non-degenerate bilinear forms on  $T_y M$  and on  $T_x M$ . On tangent vectors

$$\Omega_{(y,x)}((a, b), (a', b')) = \omega_y(a, a') - \omega_x(b, b'). \quad (3.1.3)$$

No term mixes  $a$  with  $b'$ , so the two summands are  $\Omega$ -orthogonal and  $\Omega$  is the sum of one form on each of them. Now let  $(a, b)$  be in the kernel. Taking  $b' = 0$  gives  $\omega_y(a, a') = 0$  for every  $a'$ , hence  $a = 0$ . Taking  $a' = 0$  gives  $b = 0$  the same way. Therefore  $\Omega$  is a symplectic form on the manifold  $M \times M$ . In the notation of Proposition 3.1 this manifold is  $M \times \overline{M}$ .

It is multiplicative. A composable pair of tangent vectors at  $((z, y), (y, x))$  has the form  $((c, b), (b, a))$ , with  $Tm((c, b), (b, a)) = (c, a)$ . By (3.1.3)

$$\Omega_{(z,y)}((c, b), (c', b')) + \Omega_{(y,x)}((b, a), (b', a')) = \omega_z(c, c') - \omega_x(a, a') = \Omega_{(z,x)}((c, a), (c', a')),$$

the middle point cancelling. This is (3.1.2). Hence,  $(M \times M, \Omega) \rightrightarrows M$  is a symplectic groupoid.

The structural features derived in general in the next section can be checked directly here. The units form the diagonal  $\Delta = \{(x, x)\}$ , on which  $\Omega$  vanishes,

$$\Omega_{(x,x)}((a, a), (a', a')) = \omega_x(a, a') - \omega_x(a, a') = 0.$$

This shows that  $\Delta$  is Lagrangian, of dimension  $\dim M = \frac{1}{2} \dim(M \times M)$ . The inversion  $(y, x) \mapsto (x, y)$  swaps the two factors and so pulls  $\Omega$  back to  $-\Omega$ . The source and target fibres through  $(x, x)$ , namely  $M \times \{x\}$  and  $\{x\} \times M$ , are  $\Omega$ -orthogonal.

**Example 3.2** (The cotangent groupoid of a Lie group). Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ . Its dual  $\mathfrak{g}^*$  carries the Lie–Poisson structure of Section 2.3. Example 2.4 found its cotangent algebroid to be the Lie algebroid of the action groupoid  $G \ltimes \mathfrak{g}^*$ . We show that, under the left trivialisation of  $T^*G$ , this action groupoid is the canonical cotangent symplectic groupoid.

*The groupoid.* Substituting the coadjoint action into Example 1.3 gives the structure maps of  $G \ltimes \mathfrak{g}^*$ ,

$$s(g, \mu) = \mu, \quad t(g, \mu) = \text{Ad}_g^* \mu, \quad u(\mu) = (e, \mu),$$

with  $\iota(g, \mu) = (g^{-1}, \text{Ad}_g^* \mu)$  and  $(h, \text{Ad}_g^* \mu)(g, \mu) = (hg, \mu)$ . Now we define the symplectic structure.

*The manifold.* So far the arrows are only the product  $G \times \mathfrak{g}^*$ . That product is a trivialisation of a cotangent bundle. Take  $\alpha_g \in T_g^* G$  and record it at the identity by left translation,

$$\langle \mu, \xi \rangle = \langle \alpha_g, T_e L_g \xi \rangle, \quad \xi \in \mathfrak{g}. \quad (3.1.4)$$

The map  $\alpha_g \mapsto (g, \mu)$  is a diffeomorphism  $T^*G \cong G \times \mathfrak{g}^*$  called the *left trivialisation*. From now on we write arrows as pairs  $(g, \mu)$  and use the canonical symplectic form of  $T^*G$ .

Recording  $\alpha_g$  by right translation instead gives  $\nu$  with  $\langle \nu, \xi \rangle = \langle \alpha_g, T_e R_g \xi \rangle$ . The two records are related. Conjugation  $C_{g^{-1}}(h) = g^{-1}hg$  satisfies  $L_g \circ C_{g^{-1}} = R_g$ , so  $T_e R_g = T_e L_g \circ \text{Ad}_{g^{-1}}$  and therefore

$$\langle \nu, \xi \rangle = \langle \mu, \text{Ad}_{g^{-1}} \xi \rangle = \langle \text{Ad}_g^* \mu, \xi \rangle.$$

Consequently,  $\nu = \text{Ad}_g^* \mu$ . Compare this with the structure maps above. An arrow is one covector on  $G$ . Its source and its target are the two ways of carrying that covector to the identity.

*The canonical form.* Let  $\theta^L$  be the left Maurer–Cartan form, the  $\mathfrak{g}$ -valued one-form

$$\theta_g^L(v) = T_g L_{g^{-1}} v \in \mathfrak{g}, \quad v \in T_g G,$$

which undoes left translation. In the trivialisation the tautological one-form  $\theta$  of  $T^*G$  becomes a pairing. Let  $q: T^*G \rightarrow G$  be the bundle projection. We do not call it  $\pi$ , which is reserved for Poisson structures. Let  $w$  be a tangent vector to  $T^*G$  at the point  $\alpha_g$ , written  $(g, \mu)$  in the trivialisation, and put  $v = Tq(w) \in T_g G$ . The tautological form is defined by  $\theta(w) = \langle \alpha_g, v \rangle$ . Undoing and then reapplying the left translation of  $v$  gives  $v = T_e L_g(\theta_g^L v)$ , so

$$\theta(w) = \langle \alpha_g, v \rangle = \langle \alpha_g, T_e L_g(\theta_g^L v) \rangle = \langle \mu, \theta_g^L v \rangle,$$

the last step by (3.1.4) with  $\xi = \theta_g^L v$ . The vector  $w$  was arbitrary, so at every point  $(g, \mu)$  of  $T^*G$

$$\theta_{(g, \mu)} = \langle \mu, \theta_g^L \rangle. \quad (3.1.5)$$

The symplectic form is  $\Omega = -d\theta$ .

*Multiplicativity.* We first show the analogous multiplicativity identity  $m^* \theta = \text{pr}_1^* \theta + \text{pr}_2^* \theta$ . Applying  $-d$  then gives (3.1.1) for  $\Omega$ , because  $d$  commutes with pullback.

First, we coordinatise the composable pairs. A pair  $((h, \nu), (g, \mu))$  is composable when  $s(h, \nu) = t(g, \mu)$ , that is when  $\nu = \text{Ad}_g^* \mu$ . It follows that  $\nu$  is not free and the pair is determined by  $h, g$  and  $\mu$ . The map

$$G \times G \times \mathfrak{g}^* \longrightarrow (T^*G)^{(2)}, \quad (h, g, \mu) \longmapsto ((h, \text{Ad}_g^* \mu), (g, \mu)),$$

is a diffeomorphism. In these coordinates

$$\text{pr}_1(h, g, \mu) = (h, \text{Ad}_g^* \mu), \quad \text{pr}_2(h, g, \mu) = (g, \mu), \quad m(h, g, \mu) = (hg, \mu).$$

Next, we need the differential of the multiplication of  $G$ . Write  $\text{mult}: G \times G \rightarrow G$  for it. Then

$$T_{(h, g)} \text{mult}(v, w) = T_h R_g v + T_g L_h w, \quad v \in T_h G, \quad w \in T_g G. \quad (3.1.6)$$

The differential is linear on  $T_h G \oplus T_g G$ , so it is enough to take the two summands separately. Freezing the second argument gives  $\text{mult}(\cdot, g) = R_g$ , whose differential at  $h$  is  $T_h R_g$ . Freezing the first gives  $\text{mult}(h, \cdot) = L_h$ , whose differential at  $g$  is  $T_g L_h$ .

The Maurer–Cartan form satisfies

$$\theta_{hg}^L(T_{(h, g)} \text{mult}(v, w)) = \text{Ad}_{g^{-1}}(\theta_h^L(v)) + \theta_g^L(w), \quad v \in T_h G, \quad w \in T_g G. \quad (3.1.7)$$

To prove it, put (3.1.6) into the definition of  $\theta^L$  at  $hg$ ,

$$\theta_{hg}^L(T \text{mult}(v, w)) = T_{hg} L_{(hg)^{-1}} T_h R_g v + T_{hg} L_{(hg)^{-1}} T_g L_h w.$$

The second term is  $T_g L_{g^{-1}} w = \theta_g^L(w)$ . Take the first term. Left and right translations commute, so  $L_{g^{-1}h^{-1}} \circ R_g = L_{g^{-1}} \circ R_g \circ L_{h^{-1}} = C_{g^{-1}} \circ L_{h^{-1}}$ . Differentiate this composite at  $h$ . It sends  $v$  first to  $T_h L_{h^{-1}} v = \theta_h^L(v)$ , which lies in  $T_e G = \mathfrak{g}$ , and then through  $T_e C_{g^{-1}} = \text{Ad}_{g^{-1}}$ . Hence the first term is  $\text{Ad}_{g^{-1}}(\theta_h^L(v))$ , which proves (3.1.7).

Now fix a composable pair  $(h, g, \mu)$  and a tangent vector  $(v, w, \lambda)$  at it, with  $v \in T_h G$ ,  $w \in T_g G$  and  $\lambda \in \mathfrak{g}^*$ . Multiplication sends it to  $Tm(v, w, \lambda) = (T \text{mult}(v, w), \lambda)$ . Evaluate (3.1.5) at the point  $(hg, \mu)$  and then use (3.1.7) and the definition of  $\text{Ad}^*$ ,

$$(m^* \theta)_{(hg, \mu)}(v, w, \lambda) = \langle \mu, \theta_{hg}^L(T \text{mult}(v, w)) \rangle = \langle \text{Ad}_g^* \mu, \theta_h^L(v) \rangle + \langle \mu, \theta_g^L(w) \rangle.$$

Read the two terms through (3.1.5) again. The first is  $\theta$  at the point  $(h, \text{Ad}_g^* \mu)$  evaluated on  $v$ . That point is the first arrow, so the first term is  $(\text{pr}_1^* \theta)(v, w, \lambda)$ . The second is  $\theta$  at  $(g, \mu)$  evaluated on  $w$ , so it is  $(\text{pr}_2^* \theta)(v, w, \lambda)$ . The tangent vector was arbitrary, so

$$m^* \theta = \text{pr}_1^* \theta + \text{pr}_2^* \theta$$

and  $(T^*G, \Omega) \rightrightarrows \mathfrak{g}^*$  is a symplectic groupoid. The construction is due to Coste, Dazord and Weinstein [4].

### 3.2. Structural consequences of multiplicativity

Multiplicativity is a strong condition. This section presents four consequences of it. The dimension of  $\mathcal{G}$  is fixed by the base. The unit submanifold is Lagrangian. The inversion is anti-symplectic. The source and target fibres are symplectic orthogonal complements of each other at every point. The rest of the chapter uses all four. These fundamental facts are due to Coste, Dazord and Weinstein [4]. Modern treatments are in [3, 6].

Three facts about a symplectic vector space are needed. For a linear subspace  $W \subseteq T_g\mathcal{G}$  we write

$$W^\Omega = \{ v \in T_g\mathcal{G} : \Omega_g(v, w) = 0 \text{ for all } w \in W \}$$

for its *symplectic orthogonal complement*. The first two facts are  $\dim W + \dim W^\Omega = \dim \mathcal{G}$  and  $(W^\Omega)^\Omega = W$ . The third follows from the first. An isotropic subspace satisfies  $W \subseteq W^\Omega$ . Therefore  $\dim W \leq \dim \mathcal{G} - \dim W$  and the dimension of  $W$  is at most  $\frac{1}{2} \dim \mathcal{G}$ .

Throughout the section  $(\mathcal{G} \rightrightarrows M, \Omega)$  is a symplectic groupoid.

We first record how the tangent multiplication acts on a pair with one zero entry. Nothing here uses  $\Omega$ , and the statement holds in any Lie groupoid. Let  $(g, h) \in \mathcal{G}^{(2)}$  and let  $X \in \ker T_g s$ . Then  $Ts(X) = 0$ , so the pair  $(X, 0)$  is composable. The kernel  $\ker T_g s$  is the tangent space at  $g$  to the source fibre  $\mathcal{G}_{s(g)}$ , so there is a curve  $\gamma$  inside that fibre with  $\gamma(0) = g$  and  $\gamma'(0) = X$ . Along it  $s(\gamma(\tau)) = s(g) = t(h)$ , so every pair  $(\gamma(\tau), h)$  is composable and  $m(\gamma(\tau), h) = R_h(\gamma(\tau))$ . A curve inside the target fibre  $\mathcal{G}^{t(h)}$  through  $h$  gives the mirror statement for the left translation. Differentiating both at  $\tau = 0$  gives

$$Tm(X, 0) = T_g R_h(X), \quad Tm(0, Y) = T_h L_g(Y), \quad (3.2.1)$$

for  $X \in \ker T_g s$  and  $Y \in \ker T_h t$ , where 0 is the zero tangent vector at the arrow in question.

**Lemma 3.1** (Three products). *Let  $(\mathcal{G} \rightrightarrows M, \Omega)$  be a symplectic groupoid.*

(i) *The unit submanifold  $u(M)$  is isotropic. For  $x \in M$  and  $a, b \in T_x M$ ,*

$$\Omega_{1_x}(Tu a, Tu b) = 0.$$

(ii) *The source fibre and the target fibre through an arrow are  $\Omega$ -orthogonal. For  $g \in \mathcal{G}$ ,  $X \in \ker T_g s$  and  $Y \in \ker T_g t$  one has  $\Omega_g(X, Y) = 0$ , that is  $\ker T_g t \subseteq (\ker T_g s)^\Omega$ .*

(iii) *The inversion is an anti-symplectomorphism,  $\iota^* \Omega = -\Omega$ . It carries Lagrangian submanifolds of  $(\mathcal{G}, \Omega)$  to Lagrangian submanifolds.*

*Proof.* Each part evaluates (3.1.2) on a composable pair whose product is already known. The three products are  $1_x 1_x = 1_x$ ,  $g 1_x = g$  and  $g g^{-1} = 1_{t(g)}$ .

(i). From  $s \circ u = t \circ u = \text{id}_M$  we get  $Ts(Tu a) = Tt(Tu a) = a$ , so the pair  $(Tu a, Tu a)$  is composable. The map  $x \mapsto (1_x, 1_x)$  sends  $M$  into  $\mathcal{G}^{(2)}$  and satisfies  $m \circ (u, u) = u$ . Differentiating that identity gives  $Tm(Tu a, Tu a) = Tu(a)$ , and the same for  $b$ . Hence

$$\begin{aligned} \Omega_{1_x}(Tu a, Tu b) &= \Omega_{1_x}(Tm(Tu a, Tu a), Tm(Tu b, Tu b)) \\ &= \Omega_{1_x}(Tu a, Tu b) + \Omega_{1_x}(Tu a, Tu b). \end{aligned}$$

A number equal to twice itself is zero.

(ii). Put  $x = s(g)$  and use the pair  $(g, 1_x)$ , whose product is  $g$ . The right translation  $R_{1_x}$  is the identity of the source fibre  $\mathcal{G}_x$ , so the first identity of (3.2.1) gives  $Tm(X, 0) = X$ . The

left translation  $L_g: \mathcal{G}^x \rightarrow \mathcal{G}^{t(g)}$  is a diffeomorphism of target fibres with  $L_g(1_x) = g$ , and its differential carries  $\ker T_{1_x}t$  isomorphically onto  $\ker T_g t$ . This shows that  $Y = T_{1_x}L_g(\eta)$  for a unique  $\eta \in \ker T_{1_x}t$ , and the second identity gives  $Tm(0, \eta) = Y$ . Then

$$\Omega_g(X, Y) = \Omega_g(Tm(X, 0), Tm(0, \eta)) = \Omega_g(X, 0) + \Omega_{1_x}(0, \eta) = 0.$$

(iii). Fix  $g \in \mathcal{G}$  and  $X, Y \in T_g\mathcal{G}$ . From  $t \circ \iota = s$  we get  $Tt(T\iota X) = Ts(X)$ , so the pair  $(X, T\iota X)$  is composable, and the same holds for  $(Y, T\iota Y)$ . Differentiating  $m \circ (\text{id}, \iota) = u \circ t$  gives

$$Tm(X, T\iota X) = Tu(Tt X) \in Tu(T_{t(g)}M),$$

and likewise for  $Y$ . Consequently

$$\Omega_{1_{t(g)}}(Tu Tt X, Tu Tt Y) = \Omega_g(X, Y) + \Omega_{g^{-1}}(T\iota X, T\iota Y).$$

The vectors  $Tt X$  and  $Tt Y$  lie in  $T_{t(g)}M$ , so the left side vanishes by part (i). This gives  $\iota^*\Omega = -\Omega$ . A submanifold is Lagrangian for  $\Omega$  exactly when it is Lagrangian for  $-\Omega$ , which gives the last claim.  $\square$

**Proposition 3.2** (Structure of a symplectic groupoid). *Let  $(\mathcal{G} \rightrightarrows M, \Omega)$  be a symplectic groupoid.*

(i)  $\dim \mathcal{G} = 2 \dim M$ .

(ii) *The unit submanifold  $u(M)$  is Lagrangian.*

(iii) *For every  $g \in \mathcal{G}$ ,*

$$(\ker T_g s)^\Omega = \ker T_g t, \quad (\ker T_g t)^\Omega = \ker T_g s.$$

*Proof.* Write  $n = \dim M$  and  $N = \dim \mathcal{G}$ .

Take  $g = 1_x$  in Lemma 3.1(ii), which gives  $\ker T_{1_x}t \subseteq (\ker T_{1_x}s)^\Omega$ . Both  $s$  and  $t$  are submersions, so both kernels have dimension  $N - n$ . The complement has dimension  $n$ , hence  $N - n \leq n$ , that is  $N \leq 2n$ .

By Lemma 3.1(i) the subspace  $Tu(T_x M)$  is isotropic. Its dimension is  $n$ , because  $Ts \circ Tu = \text{id}$  makes  $Tu$  injective. An isotropic subspace has dimension at most  $\frac{1}{2}N$ . Hence,  $n \leq \frac{1}{2}N$ , that is  $N \geq 2n$ . The two bounds meet and  $N = 2n$ , which gives (i).

Part (ii) follows directly. The subspace  $Tu(T_x M)$  is isotropic of dimension  $\frac{1}{2}N$  and so is Lagrangian.

For part (iii) fix  $g$ . By (i) the kernel  $\ker T_g t$  has dimension  $N - n = n$ , and  $(\ker T_g s)^\Omega$  has dimension  $N - (N - n) = n$  as well. Equal dimensions turn the inclusion of Lemma 3.1(ii) into an equality. Taking  $\Omega$ -orthogonal complements of both sides and using  $(W^\Omega)^\Omega = W$  gives the second identity.  $\square$

*Remark 3.1.* Proposition 3.1 showed that the graph of the multiplication is isotropic. Now it is Lagrangian. The composable pairs form the fibre product of two submersions. It follows that  $\mathcal{G}^{(2)}$  has codimension  $\dim M$  in  $\mathcal{G} \times \mathcal{G}$  and

$$\dim \mathcal{G}^{(2)} = 2 \dim \mathcal{G} - \dim M = \frac{3}{2} \dim \mathcal{G},$$

which is half of  $\dim(\mathcal{G} \times \mathcal{G} \times \overline{\mathcal{G}}) = 3 \dim \mathcal{G}$ . The multiplication of a symplectic groupoid is given by a Lagrangian submanifold of  $\mathcal{G} \times \mathcal{G} \times \overline{\mathcal{G}}$ .

### 3.3. The Poisson structure of the base

A symplectic groupoid determines a Poisson structure on its base. The theorem is due to Weinstein [16]. The detailed account is of Coste, Dazord and Weinstein [4]. Modern treatments are in [3, 6].

We write  $\{\cdot, \cdot\}_\Omega$  for the Poisson bracket of  $(\mathcal{G}, \Omega)$  (Example 2.1). Its Hamiltonian fields are fixed by  $\iota_{X_\phi}\Omega = d\phi$ . Hence  $X_\phi\psi = \{\psi, \phi\}$  and  $\{\phi, \psi\}_\Omega = \Omega(X_\phi, X_\psi)$ .

The whole section studies two assignments. A function on  $M$  has two pullbacks to  $\mathcal{G}$ , and each pullback has a Hamiltonian vector field,

$$f \mapsto X_{t^*f}, \quad f \mapsto X_{s^*f}, \quad f \in C^\infty(M).$$

Lemma 3.2 collects three properties of these fields, and every later result of the section applies one of them. The inversion satisfies  $\iota^*\Omega = -\Omega$  by Lemma 3.1(iii). It exchanges  $s$  with  $t$  and reverses products, so it exchanges left translation with right translation. Every statement about one assignment therefore has a mirror about the other. We state both sides and prove one of each pair.

Three pieces of bookkeeping about the tangent multiplication are needed. Nothing in them uses  $\Omega$ . First, on  $\mathcal{G}^{(2)}$  one has  $t \circ m = t \circ \text{pr}_1$  and  $s \circ m = s \circ \text{pr}_2$ , because  $t(gh) = t(g)$  and  $s(gh) = s(h)$ . Differentiating gives

$$Tt(Tm(v, v')) = Tt(v), \quad Ts(Tm(v, v')) = Ts(v') \quad (3.3.1)$$

for every composable pair  $(v, v') \in T_{(g,h)}\mathcal{G}^{(2)}$ . Second, left translation by  $g$  is a diffeomorphism between target fibres, so  $T_h L_g$  carries  $\ker T_h t$  onto  $\ker T_{gh} t$ . By (3.2.1) it does so as  $y \mapsto Tm(0, y)$ . Third, the differential  $T_{(g,h)}m: T_{(g,h)}\mathcal{G}^{(2)} \rightarrow T_{gh}\mathcal{G}$  is surjective. To see this take  $w \in T_{gh}\mathcal{G}$ . The target map is a submersion, so there is  $v \in T_g\mathcal{G}$  with  $Tt(v) = Tt(w)$ , and there is  $v' \in T_h\mathcal{G}$  with  $Tt(v') = Ts(v)$ . The pair  $(v, v')$  is composable, and  $Tt(Tm(v, v')) = Tt(v) = Tt(w)$  by (3.3.1). Therefore  $w - Tm(v, v')$  lies in  $\ker T_{gh} t$ , which is contained in the image of  $T_{(g,h)}m$  by the second point. Hence,  $w$  lies in the image as well.

**Lemma 3.2** (Pullbacks and their Hamiltonian fields). *Let  $(\mathcal{G} \rightrightarrows M, \Omega)$  be a symplectic groupoid and let  $f_1, f_2 \in C^\infty(M)$ .*

(i) *For every  $g \in \mathcal{G}$ ,*

$$\ker T_g s = \{X_{t^*f_1}(g) : f_1 \in C^\infty(M)\}, \quad \ker T_g t = \{X_{s^*f_1}(g) : f_1 \in C^\infty(M)\}.$$

(ii) *The field  $X_{t^*f_1}$  is right-invariant and the field  $X_{s^*f_1}$  is left-invariant. For every  $(g, h) \in \mathcal{G}^{(2)}$ ,*

$$T_g R_h(X_{t^*f_1}(g)) = X_{t^*f_1}(gh), \quad T_h L_g(X_{s^*f_1}(h)) = X_{s^*f_1}(gh).$$

(iii)  $\{s^*f_1, t^*f_2\}_\Omega = 0$ .

*Proof.* (i). Fix  $g$  and put  $y = t(g)$ . The differential of a target pullback is  $d(t^*f_1)_g = d(f_1)_y \circ T_g t$ . The target map is a submersion, so  $T_g t$  is onto and  $\alpha \mapsto \alpha \circ T_g t$  is an injection of  $T_y^*M$  into  $T_g^*\mathcal{G}$  with image the annihilator of  $\ker T_g t$ . Every covector at  $y$  is  $d(f_1)_y$  for some  $f_1$ , so the covectors  $d(t^*f_1)_g$  fill that annihilator. The form  $\Omega_g$  is non-degenerate. The map  $v \mapsto \Omega_g(v, \cdot)$  carries  $(\ker T_g t)^\Omega$  onto the annihilator of  $\ker T_g t$ . Since  $\Omega_g(X_{t^*f_1}(g), \cdot) = d(t^*f_1)_g$ , the vectors  $X_{t^*f_1}(g)$  fill  $(\ker T_g t)^\Omega$ , which is  $\ker T_g s$  by Proposition 3.2(iii). Exchanging  $s$  and  $t$  gives the second identity.

(ii). Put  $X = X_{t^*f_1}(g)$ . By (i) we have  $X \in \ker T_g s$ . This shows that the pair  $(X, 0)$  at  $(g, h)$  is composable and  $Tm(X, 0) = T_g R_h(X)$  by (3.2.1). Let  $w \in T_{gh}\mathcal{G}$  and write

$w = Tm(v, v')$  with  $(v, v')$  composable, which the bookkeeping above allows. Multiplicativity (3.1.2) gives

$$\Omega_{gh}(Tm(X, 0), Tm(v, v')) = \Omega_g(X, v) + \Omega_h(0, v') = d(t^*f_1)_g(v).$$

By (3.3.1) we have  $Tt(w) = Tt(v)$ , so

$$d(t^*f_1)_{gh}(w) = df_1(Ttw) = df_1(Ttv) = d(t^*f_1)_g(v).$$

Both quantities on the left depend on  $w$  alone, so the choice of the pair  $(v, v')$  plays no role. The two right-hand sides agree, hence  $\Omega_{gh}(TgR_h(X), w) = d(t^*f_1)_{gh}(w)$  for every  $w \in T_{gh}\mathcal{G}$ . This is the equation that defines  $X_{t^*f_1}(gh)$ , and the non-degeneracy of  $\Omega_{gh}$  admits only one solution. Consequently,  $TgR_h(X) = X_{t^*f_1}(gh)$ . The identity for  $X_{s^*f_1}$  is the same computation with  $s$  and  $t$  exchanged, using  $Tm(0, y) = T_hL_g(y)$  in place of  $Tm(X, 0) = TgR_h(X)$ .

(iii). By (i) the field  $X_{t^*f_2}$  takes values in  $\ker Ts$ , and the differential  $d(s^*f_1) = df_1 \circ Ts$  annihilates  $\ker Ts$ . It follows that

$$\{s^*f_1, t^*f_2\}_\Omega = X_{t^*f_2}(s^*f_1) = d(s^*f_1)(X_{t^*f_2}) = 0.$$

□

**Theorem 3.1** (Base Poisson structure). *Let  $(\mathcal{G} \rightrightarrows M, \Omega)$  be a symplectic groupoid. There is exactly one Poisson structure  $\pi$  on  $M$  for which the target map  $t: (\mathcal{G}, \Omega) \rightarrow (M, \pi)$  is a Poisson map. Its bracket is determined by*

$$t^*\{f_1, f_2\} = \{t^*f_1, t^*f_2\}_\Omega, \quad f_1, f_2 \in C^\infty(M). \quad (3.3.2)$$

The source map is a Poisson map from  $(\mathcal{G}, \Omega)$  to  $\overline{M} = (M, -\pi)$ .

*Proof.* The bracket of two target pullbacks is constant on each target fibre. Fix  $f_1, f_2 \in C^\infty(M)$  and let  $(g, h) \in \mathcal{G}^{(2)}$ . Put  $X_i = X_{t^*f_i}(g)$ , which lies in  $\ker T_g s$  by Lemma 3.2(i). By part (ii) of the same lemma and by (3.2.1),

$$X_{t^*f_i}(gh) = TgR_h(X_i) = Tm(X_i, 0),$$

so multiplicativity (3.1.2) gives

$$\{t^*f_1, t^*f_2\}_\Omega(gh) = \Omega_{gh}(Tm(X_1, 0), Tm(X_2, 0)) = \Omega_g(X_1, X_2) = \{t^*f_1, t^*f_2\}_\Omega(g).$$

Now take  $g = 1_{t(h)}$ . The pair  $(1_{t(h)}, h)$  is composable and  $1_{t(h)}h = h$ , so for every  $h \in \mathcal{G}$

$$\{t^*f_1, t^*f_2\}_\Omega(h) = \{t^*f_1, t^*f_2\}_\Omega(1_{t(h)}).$$

Put  $\{f_1, f_2\} = u^*\{t^*f_1, t^*f_2\}_\Omega$ , a smooth function on  $M$ . Its pullback along  $t$  takes at  $h$  the value  $\{t^*f_1, t^*f_2\}_\Omega(1_{t(h)})$ . This is the identity (3.3.2).

*The bracket is Poisson.* The target map is surjective, so  $t^*$  is injective. Bilinearity is clear from (3.3.2). For the Jacobi identity apply (3.3.2) twice,

$$t^*\{\{f_1, f_2\}, f_3\} = \{t^*\{f_1, f_2\}, t^*f_3\}_\Omega = \{\{t^*f_1, t^*f_2\}_\Omega, t^*f_3\}_\Omega,$$

and add the two identities obtained by cycling  $f_1, f_2$  and  $f_3$ . The right side becomes the Jacobi expression of  $\{\cdot, \cdot\}_\Omega$  on  $t^*f_1, t^*f_2$  and  $t^*f_3$ , which is zero. Hence  $t^*$  of the Jacobi expression on  $M$  is zero, and it vanishes because  $t^*$  is injective. Antisymmetry and the Leibniz rule follow in the same way from one application of (3.3.2) instead of two.

*Uniqueness.* A Poisson structure for which  $t$  is a Poisson map satisfies (3.3.2), and that determines every bracket  $\{f_1, f_2\}$  because  $t^*$  is injective.

*The source map.* Write  $s = t \circ \iota$  and use  $\iota^*\Omega = -\Omega$  from Lemma 3.1(iii). A symplectic form and its negative have brackets differing by a sign, so  $\{\iota^*A, \iota^*B\}_\Omega = -\iota^*\{A, B\}_\Omega$  for  $A, B \in C^\infty(\mathcal{G})$ . Hence

$$s^*\{f_1, f_2\} = \iota^*t^*\{f_1, f_2\} = \iota^*\{t^*f_1, t^*f_2\}_\Omega = -\{s^*f_1, s^*f_2\}_\Omega,$$

which is (3.3.2) for  $s$  and  $-\pi$ . □

We call  $\pi$  the induced structure. Everything later in this chapter and in Chapter 4 rests on the identity (3.3.2).

*Remark 3.2* (Conventions). Sources differ over which structure map is the Poisson one. Weinstein states the theorem with the source map Poisson and the target map anti-Poisson [16], and the lectures of Crainic and Fernandes do the same [5]. We use the opposite orientation. It matches the conventions of Remark 2.2 and of the cotangent groupoid in Example 3.2. A reader comparing statements with the literature should check the orientation first.

**Definition 3.2** (Symplectic realisation). Let  $(M, \pi)$  be a Poisson manifold. A *symplectic realisation* of  $(M, \pi)$  is a symplectic manifold together with a surjective submersion onto  $M$  which is a Poisson map.

By Theorem 3.1 the target map of a symplectic groupoid is a symplectic realisation of  $(M, \pi)$  and the source map is one of  $\overline{M}$ ,

$$M \xleftarrow{t} \mathcal{G} \xrightarrow{s} \overline{M}.$$

The fibres of the two maps are symplectic orthogonal complements of each other by Proposition 3.2(iii). A pair of realisations of opposite sign with this orthogonality is called a *dual pair*. Weinstein obtained Theorem 3.1 from the theory of dual pairs [3, 16]. Chapter 4 uses the two structure maps to read Lie–Poisson reduction.

**Corollary 3.1** (Lifted dynamics). *Let  $(\mathcal{G} \rightrightarrows M, \Omega)$  be a symplectic groupoid, let  $\pi$  be the induced structure, let  $H \in C^\infty(M)$  and let  $\varphi_\tau$  be the flow of  $X_H$  on  $(M, \pi)$ .*

- (i) *The field  $X_{t^*H}$  is tangent to the source fibres and*

$$Tt(X_{t^*H}) = X_H \circ t.$$

*Its flow  $\tilde{\varphi}_\tau$  satisfies  $s \circ \tilde{\varphi}_\tau = s$  and  $t \circ \tilde{\varphi}_\tau = \varphi_\tau \circ t$  wherever it is defined.*

- (ii) *The field  $X_{s^*H}$  is tangent to the target fibres and*

$$Ts(X_{s^*H}) = -X_H \circ s.$$

*Its flow  $\hat{\varphi}_\tau$  satisfies  $t \circ \hat{\varphi}_\tau = t$  and  $s \circ \hat{\varphi}_\tau = \varphi_{-\tau} \circ s$  wherever it is defined.*

*Proof.* The two tangency statements are Lemma 3.2(i). For the projection in (i) test against an arbitrary  $f_1 \in C^\infty(M)$ . Use  $d(t^*f_1) = df_1 \circ Tt$ ,

$$\langle df_1, Tt X_{t^*H} \rangle = X_{t^*H}(t^*f_1) = \{t^*f_1, t^*H\}_\Omega = t^*\{f_1, H\} = \langle df_1, X_H \rangle \circ t,$$

by (3.3.2) and  $\{f_1, H\} = X_H f_1$  (Section 2.2.1). The covector  $df_1$  is arbitrary, so the two tangent vectors agree. In (ii) the same computation runs with  $s$  in place of  $t$ . The middle step  $\{s^*f_1, s^*H\}_\Omega = -s^*\{f_1, H\}$  carries a minus sign, because  $s$  is a Poisson map onto  $\overline{M}$ .

For the flows, differentiate along them. In (i) the curve  $\tau \mapsto s(\tilde{\varphi}_\tau(g))$  has velocity  $Ts(X_{t^*H}) = 0$ , so it stays at  $s(g)$ . The curve  $\tau \mapsto t(\tilde{\varphi}_\tau(g))$  has velocity  $X_H$  at its current point. Therefore it is the integral curve of  $X_H$  through  $t(g)$ , and integral curves through a point are unique. In (ii) the same argument applies with  $-X_H$  in place of  $X_H$ .  $\square$

Start the flow of  $X_{s^*H}$  at the units. By (ii) the targets stay fixed and the sources move along the flow of  $H$ . This shows that the exact solution of the Hamiltonian system appears on  $\mathcal{G}$  as a curve of arrows through the units. Section 3.4 turns this curve into so-called bisections. Chapter 4 reads the same curve on  $T^*G$  as the unreduced motion of a left-invariant Hamiltonian.

**Proposition 3.3** (Leaves, orbits and Casimirs). *Let  $(\mathcal{G} \rightrightarrows M, \Omega)$  be a symplectic groupoid and let  $\pi$  be the induced structure.*

(i) For every  $g \in \mathcal{G}$ ,

$$\operatorname{im} \pi_{t(g)}^\sharp = T_g t(\ker T_g s).$$

(ii) If the source fibres of  $\mathcal{G}$  are connected, the symplectic leaves of  $(M, \pi)$  are the orbits of  $\mathcal{G}$ .

(iii) If the source fibres of  $\mathcal{G}$  are connected, a function  $C \in C^\infty(M)$  is a Casimir of  $\pi$  if and only if  $s^*C = t^*C$ .

*Proof.* (i). By Lemma 3.2(i) the space  $\ker T_g s$  consists of the values  $X_{t^*f_1}(g)$  with  $f_1 \in C^\infty(M)$ . By Corollary 3.1(i),

$$T_g t(X_{t^*f_1}(g)) = X_{f_1}(t(g)) = \pi^\sharp(d(f_1)_{t(g)}).$$

Every covector at  $t(g)$  is  $d(f_1)_{t(g)}$  for some  $f_1$ . Consequently, as  $f_1$  varies the right side fills  $\operatorname{im} \pi_{t(g)}^\sharp$ .

(ii). Recall  $\mathcal{O}_x = t(\mathcal{G}_x)$  from Section 1.2.2. We use the immersed-orbit theorem for Lie groupoids: the groupoid orbit is an immersed submanifold whose tangent space at  $t(g)$  is  $T_g t(\ker T_g s)$ . By part (i), this tangent space is precisely the characteristic distribution  $\operatorname{im} \pi^\sharp$  along the orbit. If the source fibres are connected,  $\mathcal{O}_x$  is a connected integral manifold of the characteristic distribution and is contained in the maximal symplectic leaf through  $x$ .

For the reverse inclusion take  $H \in C^\infty(M)$  with flow  $\varphi_\tau$  on  $M$ . By Corollary 3.1(ii) the arrow  $\widehat{\varphi}_\tau(1_x)$  has target  $t(1_x) = x$  and source  $\varphi_{-\tau}(x)$ , so  $\varphi_{-\tau}(x)$  lies in  $\mathcal{O}_x$  for small  $\tau$ . Apply this at an arbitrary point of the curve. For every  $\tau_0$  in the interval of definition, all points  $\varphi_{-\tau}(x)$  with  $\tau$  near  $\tau_0$  lie in the orbit of  $\varphi_{-\tau_0}(x)$ . Orbits are disjoint, so the set of times  $\tau$  with  $\varphi_{-\tau}(x) \in \mathcal{O}_x$  is open and closed in that interval. It contains  $\tau = 0$ , so the whole curve stays in  $\mathcal{O}_x$ . Two points of one leaf are joined by a chain of Hamiltonian flows by Theorem 2.1(i), and each flow keeps the point in its orbit. It follows that the leaf through  $x$  is contained in  $\mathcal{O}_x$ .

(iii). Let  $C$  be a Casimir. Then  $\{C, f_1\} = 0$  for every  $f_1$ , so  $C$  is constant along every Hamiltonian flow, and by Theorem 2.1(i) it is constant on each leaf. By (ii) it is constant on each orbit. For  $g \in \mathcal{G}$  the points  $s(g)$  and  $t(g)$  lie in one orbit, so  $C(s(g)) = C(t(g))$ .

Conversely let  $s^*C = t^*C$ . Then  $X_{s^*C} = X_{t^*C}$ , and by Lemma 3.2(i) this field lies in  $\ker Tt$  and in  $\ker Ts$ . Its image under  $Tt$  is  $X_C \circ t$  by Corollary 3.1(i), and that image is zero. The target map is onto, so  $X_C = 0$  and  $C$  is a Casimir by Section 2.2.2.  $\square$

The hypothesis in parts (ii) and (iii) cannot be dropped for an arbitrary groupoid. It can, however, be achieved without changing the Lie algebroid or the induced Poisson tensor: the union of the connected components of the source fibres that meet the unit section is an open subgroupoid over the same base, and it has the same Lie algebroid [13]. Restricting  $\Omega$  makes it a symplectic groupoid again, and the induced structure does not change, because it is computed on the units. The inversion carries source fibres onto target fibres, so the hypothesis holds for the target fibres as well. In Chapter 4 parts (ii) and (iii) identify the symplectic leaves and the Casimirs of  $\mathfrak{g}^*$ .

**Example 3.3** (The base of the pair groupoid). Let  $(M, \omega)$  be a symplectic manifold and take  $(M \times M, \Omega) \rightrightarrows M$  of Example 3.1, with  $t = \operatorname{pr}_1$ ,  $s = \operatorname{pr}_2$  and  $\Omega = \operatorname{pr}_1^* \omega - \operatorname{pr}_2^* \omega$ . Solve  $\iota_X \Omega = d(t^*f_1)$  for  $X = (X_1, X_2)$ . By (3.1.3) this reads

$$\omega(X_1, a) - \omega(X_2, b) = df_1(a) \quad \text{for all } (a, b),$$

so  $X_1 = X_{f_1}$  and  $X_2 = 0$ . Hence  $X_{t^*f_1} = (X_{f_1}, 0)$  and

$$\{t^*f_1, t^*f_2\}_\Omega = \Omega((X_{f_1}, 0), (X_{f_2}, 0)) = \omega(X_{f_1}, X_{f_2}) = t^*\{f_1, f_2\}_\omega,$$

with  $\{\cdot, \cdot\}_\omega$  the bracket of  $(M, \omega)$  (Example 2.1). By (3.3.2) the structure induced on the base is  $\{\cdot, \cdot\}_\omega$  itself.

### 3.4. Bisections

The algebroid of Chapter 1 carries infinitesimal motions of the base. Here we introduce the finite ones. A *bisection* is a choice of one arrow out of each point of the base. It moves the base by a diffeomorphism, and when it is Lagrangian it moves the base by a Poisson diffeomorphism. Bisections of Lie groupoids are treated by Mackenzie [9]. Their Lagrangian version on a symplectic groupoid is due to Coste, Dazord and Weinstein [4].

**Definition 3.3.** A *bisection* of a Lie groupoid  $\mathcal{G} \rightrightarrows M$  is a submanifold  $B \subseteq \mathcal{G}$  on which both the source and the target restrict to diffeomorphisms onto  $M$ . A bisection of a symplectic groupoid is a *Lagrangian bisection* if in addition  $B$  is a Lagrangian submanifold of  $(\mathcal{G}, \Omega)$ .

A bisection meets each source fibre once and each target fibre once. The unit section  $u(M)$  is the bisection of identity arrows. Every bisection induces a diffeomorphism of the base,

$$\psi_B = t|_B \circ (s|_B)^{-1}: M \longrightarrow M, \quad (3.4.1)$$

which carries a point  $x$  to the target of the arrow of  $B$  whose source is  $x$ .

**Lemma 3.3.** Let  $B$  be a bisection and put  $\sigma = (s|_B)^{-1}: M \rightarrow \mathcal{G}$ . Then  $\sigma$  is a smooth section of  $s$  with image  $B$ , and  $t \circ \sigma = \psi_B$  is a diffeomorphism. Conversely, let  $\sigma: M \rightarrow \mathcal{G}$  be a smooth section of  $s$  for which  $t \circ \sigma$  is a diffeomorphism. Then  $\sigma(M)$  is a bisection.

*Proof.* The first part is a restatement of the definition together with (3.4.1). For the converse, a section of a surjective submersion is an embedding, so  $B = \sigma(M)$  is a submanifold and  $s|_B$  is the inverse of  $\sigma$ , hence a diffeomorphism onto  $M$ . Then  $t|_B$  factors as  $(t \circ \sigma) \circ s|_B$ , a composite of two diffeomorphisms.  $\square$

We read both running examples through the lemma.

**Example 3.4** (Bisections of the pair groupoid). Take  $(M \times M, \Omega) \rightrightarrows M$  of Example 3.1, with  $s(y, x) = x$  and  $t(y, x) = y$ . A section of  $s$  is a map  $x \mapsto (\Phi(x), x)$  for a smooth  $\Phi: M \rightarrow M$ , and  $t$  composed with it is  $\Phi$  itself. By Lemma 3.3 the bisections are exactly the graphs

$$\text{Gr}(\Phi) = \{(\Phi(x), x) : x \in M\}$$

of diffeomorphisms  $\Phi$  of  $M$ , with  $\psi_{\text{Gr}(\Phi)} = \Phi$ . The bisection group is  $\text{Diff}(M)$ .

Such a graph is Lagrangian exactly when  $\Phi$  is a symplectomorphism. Parametrise it by  $\sigma(x) = (\Phi(x), x)$  and compute

$$\sigma^* \Omega = (\text{pr}_1 \circ \sigma)^* \omega - (\text{pr}_2 \circ \sigma)^* \omega = \Phi^* \omega - \omega.$$

The graph has dimension  $\dim M$ , which is half of  $\dim(M \times M)$ . Hence it is Lagrangian if and only if  $\sigma^* \Omega = 0$ . That is if and only if  $\Phi^* \omega = \omega$ . The Lagrangian bisections of the pair groupoid are the canonical transformations of  $(M, \omega)$ .

**Example 3.5** (Bisections of the cotangent groupoid). Take  $T^*G \rightrightarrows \mathfrak{g}^*$  of Example 3.2, with  $s(g, \mu) = \mu$  and  $t(g, \mu) = \text{Ad}_g^* \mu$ . A section of  $s$  is a map  $\mu \mapsto (b(\mu), \mu)$  for a smooth  $b: \mathfrak{g}^* \rightarrow G$ . Therefore the sections of the source are just the smooth maps into the group. By Lemma 3.3 such a section is a bisection exactly when

$$\psi_B(\mu) = \text{Ad}_{b(\mu)}^* \mu \quad (3.4.2)$$

is a diffeomorphism of  $\mathfrak{g}^*$ . Any  $b$  gives a section of the source, and the whole content of the definition sits in (3.4.2). It moves each  $\mu$  inside its  $G$ -coadjoint orbit. If  $G$  is connected, or if  $b$  takes values in the identity component, then  $\psi_B$  preserves every symplectic leaf and hence every Casimir. For disconnected  $G$ , preservation of full  $G$ -orbits alone does not imply preservation of every Casimir.

**Proposition 3.4.** *The bisections of  $\mathcal{G}$  form a group under the product*

$$B \cdot B' = \{ b b' : b \in B, b' \in B', s(b) = t(b') \},$$

*with unit  $u(M)$  and with inverse  $\iota(B)$ . The map  $B \mapsto \psi_B$  is a homomorphism into  $\text{Diff}(M)$ .*

*Proof.* Let  $\sigma$  and  $\sigma'$  be the sections of Lemma 3.3. For  $x \in M$  the arrow  $\sigma'(x)$  has target  $\psi_{B'}(x)$ , and  $\sigma(\psi_{B'}(x))$  has source  $\psi_{B'}(x)$ . This shows that the two are composable and

$$\tau(x) = \sigma(\psi_{B'}(x)) \sigma'(x)$$

is defined and smooth. Its source is  $x$ , so  $\tau$  is a section of  $s$  with image  $B \cdot B'$ . Its target is  $\psi_B(\psi_{B'}(x))$ . That composite is a diffeomorphism. Consequently,  $B \cdot B'$  is a bisection by Lemma 3.3, and  $\psi_{B \cdot B'} = \psi_B \circ \psi_{B'}$ .

The unit section is a bisection with  $\psi_{u(M)} = \text{id}_M$ . Relations  $b u(s(b)) = b$  and  $u(t(b)) b = b$  make it a unit for the product. For the inverse,  $s(b^{-1}) = t(b)$  and  $t(b^{-1}) = s(b)$ . It follows that  $s$  and  $t$  restricted to  $\iota(B)$  are  $t|_B$  and  $s|_B$  read through the inversion. Both are diffeomorphisms, so  $\iota(B)$  is a bisection, and  $\psi_{\iota(B)} = s|_B \circ (t|_B)^{-1} = \psi_B^{-1}$ . Finally take  $b, b' \in B$  with  $s(b) = t((b')^{-1}) = s(b')$ . Injectivity of  $s|_B$  forces  $b = b'$ , so  $B \cdot \iota(B) = \{ b b^{-1} \} = u(M)$ . The same argument with  $t|_B$  gives  $\iota(B) \cdot B = u(M)$ .  $\square$

A bisection also acts on the groupoid itself. This is how it reaches the base.

**Proposition 3.5** (Left translation). *Let  $B$  be a bisection with section  $\sigma$ . The map*

$$A_B: \mathcal{G} \longrightarrow \mathcal{G}, \quad A_B(g) = \sigma(t(g)) g,$$

*is a diffeomorphism of  $\mathcal{G}$  with inverse  $A_{\iota(B)}$ , and it satisfies*

$$s \circ A_B = s, \quad t \circ A_B = \psi_B \circ t.$$

*If  $\mathcal{G}$  is symplectic and  $B$  is Lagrangian, then  $A_B$  is a symplectomorphism of  $(\mathcal{G}, \Omega)$ .*

*Proof.* The product is defined because  $\sigma(t(g))$  has source  $t(g)$ . It is smooth as a composite of smooth maps. That  $A_{\iota(B)}$  inverts  $A_B$  follows from Proposition 3.4. Multiplication does not change the source of the right factor or the target of the left factor, so  $s(A_B(g)) = s(g)$  and  $t(A_B(g)) = t(\sigma(t(g))) = \psi_B(t(g))$ .

For the last claim write  $A_B = m \circ \rho$ , where  $\rho(g) = (\sigma(t(g)), g)$  is a map into  $\mathcal{G}^{(2)}$ . Multiplicativity (3.1.1) gives

$$A_B^* \Omega = \rho^* m^* \Omega = (\text{pr}_1 \circ \rho)^* \Omega + (\text{pr}_2 \circ \rho)^* \Omega = (\sigma \circ t)^* \Omega + \Omega.$$

Now  $(\sigma \circ t)^* \Omega = t^* \sigma^* \Omega$ , and  $\sigma^* \Omega = 0$  because  $\sigma$  parametrises  $B$  and  $B$  is Lagrangian. Hence  $A_B^* \Omega = \Omega$ .  $\square$

**Proposition 3.6.** *Let  $(\mathcal{G} \rightrightarrows M, \Omega)$  be a symplectic groupoid and give  $M$  the Poisson structure  $\pi$  of Theorem 3.1. If  $B$  is a Lagrangian bisection then  $\psi_B$  is a Poisson diffeomorphism of  $(M, \pi)$ . The Lagrangian bisections form a subgroup of the bisection group, and  $B \mapsto \psi_B$  restricts to a homomorphism of that subgroup into the Poisson diffeomorphisms of  $(M, \pi)$ .*

*Proof.* By Proposition 3.5 the map  $A_B$  is a symplectomorphism of  $(\mathcal{G}, \Omega)$ , hence a Poisson map for the bracket of  $\Omega$ . It covers  $\psi_B$  through the target, which is Poisson by Theorem 3.1. Thus the composite  $\psi_B \circ t = t \circ A_B$  is Poisson. The target is also a surjective submersion, so  $t^*$  is injective. This forces  $\psi_B$  itself to be Poisson.

For the subgroup, note that  $B \cdot B' = A_B(B')$ . If  $B$  and  $B'$  are Lagrangian then  $A_B$  is a symplectomorphism, so  $A_B(B')$  is Lagrangian. The inversion is anti-symplectic by Lemma 3.1(iii). A form and its negative have the same Lagrangian submanifolds, so  $\iota(B)$  is Lagrangian too. The homomorphism property is Proposition 3.4.  $\square$

A bisection does not need to be Lagrangian to respect the degeneracy of  $\pi$ . By (3.4.1) the points  $x$  and  $\psi_B(x)$  are the two ends of one arrow, so they lie in one orbit of  $\mathcal{G}$ . Let the source fibres of  $\mathcal{G}$  be connected. Then  $\psi_B$  preserves each symplectic leaf by Proposition 3.3(ii), and  $C \circ \psi_B = C$  for every Casimir  $C$  by part (iii) of the same proposition. Example 3.5 shows this for the cotangent groupoid.

*Remark 3.3.* The converse fails in general. On an arbitrary symplectic groupoid not every Poisson diffeomorphism of the base is induced by a Lagrangian bisection. For the pair groupoid, every bisection is a graph and Example 3.4 shows that canonical transformations of the symplectic base correspond exactly to Lagrangian bisections. This statement does not imply that every symplectomorphism lifts to a bisection of an arbitrary integration of the same Poisson manifold. A Hamiltonian flow nevertheless admits the lift described in the following proposition, on the stated domains of definition.

**Proposition 3.7.** *Let  $(\mathcal{G} \rightrightarrows M, \Omega)$  be a symplectic groupoid, let  $\pi$  be the induced structure, let  $H \in C^\infty(M)$  have Hamiltonian flow  $\varphi_\tau$  on  $(M, \pi)$ , and let  $\widehat{\varphi}_\tau$  be the flow on  $\mathcal{G}$  of the Hamiltonian vector field  $X_{s^*H}$ . Fix  $\tau$  such that  $\varphi_\tau$  and  $\varphi_{-\tau}$  are both defined on all of  $M$ , and such that  $\widehat{\varphi}_\tau$  is defined on all of  $u(M)$ . Then the set*

$$B_\tau := \widehat{\varphi}_\tau(u(M))$$

*is a Lagrangian bisection of  $\mathcal{G}$ , with  $B_0 = u(M)$  and  $\psi_{B_\tau} = \varphi_\tau$ .*

*Proof.*  $B_\tau$  is Lagrangian. The map  $\widehat{\varphi}_\tau$  is the flow of a Hamiltonian vector field on  $(\mathcal{G}, \Omega)$ , so it is a symplectomorphism where it is defined. The unit section is Lagrangian by Proposition 3.2, and a symplectomorphism carries Lagrangian submanifolds to Lagrangian submanifolds.

$B_\tau$  is a bisection with base map  $\varphi_\tau$ . By Corollary 3.1(ii) the flow satisfies  $t \circ \widehat{\varphi}_\tau = t$  and  $s \circ \widehat{\varphi}_\tau = \varphi_{-\tau} \circ s$ . Hence  $t \circ (\widehat{\varphi}_\tau \circ u) = t \circ u = \text{id}_M$  and  $s \circ (\widehat{\varphi}_\tau \circ u) = \varphi_{-\tau}$ . Now put  $\sigma_\tau = (\widehat{\varphi}_\tau \circ u) \circ \varphi_\tau$ . Then  $s \circ \sigma_\tau = \varphi_{-\tau} \circ \varphi_\tau = \text{id}_M$ , so  $\sigma_\tau$  is a smooth section of  $s$ , and  $t \circ \sigma_\tau = \varphi_\tau$  is a diffeomorphism. By Lemma 3.3 its image is a bisection with base map  $\varphi_\tau$ . That image is  $B_\tau$ , because  $\varphi_\tau$  is a bijection of  $M$ .  $\square$

For the pair groupoid the proposition returns  $B_\tau = \text{Gr}(\varphi_\tau)$ , so a complete Hamiltonian flow is a one-parameter subgroup of the group of canonical transformations; without completeness the statement is local on the common domains of definition. For the cotangent groupoid it is equally concrete. By Example 3.5 the bisection  $B_\tau$  is a smooth map  $b_\tau: \mathfrak{g}^* \rightarrow G$  with  $b_0 \equiv e$ , and  $\psi_{B_\tau} = \varphi_\tau$  reads

$$\text{Ad}_{b_\tau(\mu)}^* \mu = \varphi_\tau(\mu). \quad (3.4.3)$$

Whenever the hypotheses of Proposition 3.7 hold, the exact Hamiltonian flow on  $\mathfrak{g}^*$  therefore moves each momentum by a coadjoint transformation, with the group element chosen smoothly. The corresponding statement is local when the flows are only locally defined. Conservation of every Casimir follows from the Hamiltonian character of the flow; in the present groupoid description it is also visible from (3.4.3).

*Remark 3.4* (Generating functions). The Lagrangian neighbourhood theorem identifies a neighbourhood of the unit section with a neighbourhood of the zero section in  $T^*M$  [2]. Locally, and subject to the relevant transversality condition, a Lagrangian submanifold which is a section of the cotangent projection is the graph of a closed one-form. On a contractible open set this form can be written as  $dS$ , with  $S$  determined up to an additive constant. To obtain a local groupoid bisection one must additionally check that the source and target restrictions are local diffeomorphisms. Thus generating functions describe Lagrangian bisections locally rather than globally in general. For  $T^*G$ , the graph of  $dS$  for a function on an open subset of  $G$  is Lagrangian, but it still has to satisfy the corresponding bisection transversality conditions.

### 3.5. The Lie algebroid of a symplectic groupoid

Chapter 1 defines a Lie algebroid of a Lie groupoid. Chapter 2 defines the cotangent Lie algebroid of a Poisson manifold (Theorem 2.3). Theorem 3.1 shows that every symplectic groupoid admits a Poisson structure on its base. This section shows that the three constructions close on each other. The Lie algebroid of a symplectic groupoid is the cotangent algebroid of its base. The result is due to Coste, Dazord and Weinstein [4]. In the lectures of Crainic and Fernandes it is stated as one theorem together with Theorem 3.1 [5].

The identification rests on linear algebra at the units. Fix  $x \in M$ . The fibre  $A(\mathcal{G})_x = \ker T_{1_x}t$  and the subspace  $T_{1_x}u(M) = Tu(T_xM)$  are transverse in  $T_{1_x}\mathcal{G}$ . That is because a vector in both is  $Tu(a)$  with  $a = Tt(Tu(a)) = 0$ , and the dimensions add to  $\dim M + \dim M = \dim \mathcal{G}$  by Proposition 3.2. Now suppose  $v \in A(\mathcal{G})_x$  satisfies  $\Omega_{1_x}(v, Tu(a)) = 0$  for every  $a \in T_xM$ . That means that  $v$  lies in the symplectic orthogonal of  $Tu(T_xM)$ . The subspace  $Tu(T_xM)$  is Lagrangian by the same proposition, and a Lagrangian subspace equals its own symplectic orthogonal. Therefore  $v$  lies both in  $Tu(T_xM)$  and in  $A(\mathcal{G})_x$ . The transversality gives  $v = 0$ . The following pairing is therefore injective, and the two spaces have equal dimension. Hence

$$v \longmapsto -\Omega_{1_x}(v, Tu(\cdot))$$

is a linear isomorphism from  $A(\mathcal{G})_x$  onto  $T_x^*M$ . The sign is a choice, fixed so that the anchors match in the theorem.

**Theorem 3.2** (Coste–Dazord–Weinstein). *Let  $(\mathcal{G} \rightrightarrows M, \Omega)$  be a symplectic groupoid and let  $\pi$  be the induced structure on  $M$ . The bundle map*

$$\Psi: A(\mathcal{G}) \longrightarrow T^*M, \quad \langle \Psi(v), a \rangle = -\Omega_{1_x}(v, Tu(a)) \quad (v \in A(\mathcal{G})_x, a \in T_xM),$$

*is an isomorphism of Lie algebroids onto the cotangent algebroid of  $(M, \pi)$ . It intertwines the anchors,  $\pi^\sharp \circ \Psi = \rho$ , it carries the bracket of  $A(\mathcal{G})$  to the Koszul bracket, and it satisfies*

$$\Psi(X_{s^*f_1}(1_x)) = -d(f_1)_x, \quad f_1 \in C^\infty(M). \quad (3.5.1)$$

*Proof.*  $\Psi$  is an isomorphism of vector bundles. The paragraph before the theorem gives the fibrewise isomorphism, and  $\Psi$  is smooth because  $\Omega$  and  $Tu$  are.

*The identity (3.5.1).* The vector  $X_{s^*f_1}(1_x)$  lies in  $\ker T_{1_x}t = A(\mathcal{G})_x$  by Lemma 3.2(i). For  $a \in T_xM$ ,

$$\begin{aligned} \langle \Psi(X_{s^*f_1}(1_x)), a \rangle &= -\Omega_{1_x}(X_{s^*f_1}(1_x), Tu(a)) \\ &= -d(s^*f_1)(Tu(a)) \\ &= -df_1(TsTu(a)) = -df_1(a). \end{aligned}$$

because  $\iota_{X_{s^*f_1}}\Omega = d(s^*f_1)$  and  $s \circ u = \text{id}_M$ .

*The spanning sections.* For  $f_1 \in C^\infty(M)$  put

$$\alpha_{f_1} = \Psi^{-1}(df_1) = -X_{s^*f_1}|_{u(M)} \in \Gamma(A(\mathcal{G})).$$

The differentials  $d(f_1)_x$  fill  $T_x^*M$  as  $f_1$  varies, so the values  $\alpha_{f_1}(x)$  fill  $A(\mathcal{G})_x$ . The field  $-X_{s^*f_1}$  takes values in  $\ker Tt$  and is left-invariant, by Lemma 3.2(i) and (ii). A left-invariant field is determined by its values at the units (Section 1.3.1). This shows that the left-invariant extension of  $\alpha_{f_1}$  is

$$\overleftarrow{\alpha}_{f_1} = -X_{s^*f_1}.$$

*The anchors.* By Corollary 3.1(ii) and by  $X_{f_1} = \pi^\sharp df_1$  (Section 2.2.1),

$$\rho(\alpha_{f_1})(x) = Ts(-X_{s^*f_1}(1_x)) = X_{f_1}(x) = \pi^\sharp(d(f_1)_x) = \pi^\sharp(\Psi \alpha_{f_1})(x).$$

The values  $\alpha_{f_1}(x)$  fill every fibre,  $\rho$  and  $\pi^\sharp \circ \Psi$  are bundle maps, so  $\rho = \pi^\sharp \circ \Psi$ .

*The brackets on the spanning sections.* The bracket of  $A(\mathcal{G})$  restricts the bracket of left-invariant fields to the units (Section 1.3.1), so

$$[\alpha_{f_1}, \alpha_{f_2}] = [X_{s^*f_1}, X_{s^*f_2}]|_{u(M)} = -X_{\{s^*f_1, s^*f_2\}_\Omega}|_{u(M)} = X_{s^*\{f_1, f_2\}}|_{u(M)} = -\alpha_{\{f_1, f_2\}},$$

where the second step is Proposition 2.1(ii) on  $(\mathcal{G}, \Omega)$  and the third is the anti-Poisson property of  $s$  from Theorem 3.1. Applying  $\Psi$  and using (2.4.1),

$$\Psi([\alpha_{f_1}, \alpha_{f_2}]) = -d\{f_1, f_2\} = [df_1, df_2]_\pi = [\Psi \alpha_{f_1}, \Psi \alpha_{f_2}]_\pi.$$

*The brackets on all sections.* Transport the Koszul bracket through  $\Psi$ . That is put  $[\alpha, \beta]' = \Psi^{-1}[\Psi \alpha, \Psi \beta]_\pi$ . It is antisymmetric and satisfies the Jacobi identity because the Koszul bracket does. It satisfies the Leibniz rule with the anchor  $\rho$ , since by Theorem 2.3 and by the anchor identity

$$[\alpha, f_1 \beta]' = \Psi^{-1}\left(f_1 [\Psi \alpha, \Psi \beta]_\pi + (\pi^\sharp \Psi \alpha)(f_1) \Psi \beta\right) = f_1 [\alpha, \beta]' + \rho(\alpha)(f_1) \beta.$$

The brackets  $[\cdot, \cdot]$  and  $[\cdot, \cdot]'$  agree on the sections  $\alpha_{f_1}$ . Around any point choose  $c_1, \dots, c_n \in C^\infty(M)$  so that  $dc_1, \dots, dc_n$  span  $T^*M$ . Then  $\alpha_{c_1}, \dots, \alpha_{c_n}$  frame  $A(\mathcal{G})$ , and every section expands in this frame with smooth coefficients. Both brackets are antisymmetric and expand through the same Leibniz rule with the same anchor. In the expansions the coefficients multiply brackets of the  $\alpha_{c_i}$ . They agree, and the derivative terms contain only the anchor. Consequently the two brackets agree on all sections, and  $\Psi$  carries  $[\cdot, \cdot]$  to  $[\cdot, \cdot]_\pi$ .  $\square$

*Remark 3.5 (Conventions).* The minus sign in  $\Psi$  records that two choices were made independently of each other. Section 1.3.1 builds  $A(\mathcal{G})$  from  $\ker(Tt)$  with left-invariant fields, and Theorem 3.1 takes the target map as the Poisson one, within the sign conventions of Remark 2.2 and the orientation of Remark 3.2. Aligned choices would remove the sign. With  $\ker(Tt)$  and a Poisson source map the proof gives  $\Psi^{-1}(df_1) = X_{s^*f_1}|_{u(M)}$ , and with  $\ker(Ts)$ , right-invariant fields and a Poisson target map it gives  $\Psi^{-1}(df_1) = X_{t^*f_1}|_{u(M)}$ , in both cases without a sign. Sources differ along both axes, so a reader comparing with the literature should check the two together. The same mismatch produced the minus sign in the map  $\Phi$  of Example 2.4.

The invariants of the two algebroids also match. Chapter 1 shows that the isotropy Lie algebras of  $A(\mathcal{G})$  agree with those of  $\mathcal{G}$ , and that their orbits agree when the source fibres are connected. For the cotangent algebroid, the anchor is  $\pi^\sharp$ , so its orbits are the symplectic leaves and its isotropy at  $x$  is  $\ker \pi_x^\sharp$ . Under  $\Psi$  these statements agree with Proposition 3.3(ii). In particular the Lie algebra of the vertex group  $\mathcal{G}_x^x$  is carried by  $\Psi$  onto  $\ker \pi_x^\sharp$ . It is the space of covectors at  $x$  that vanish on the tangent space of the leaf, with the bracket induced by the Koszul bracket. In general, this bracket need not be described solely by the vector-space structure of  $\ker \pi_x^\sharp$ .

**Example 3.6** (The algebroid of the pair groupoid). Take  $(M \times M, \Omega) \rightrightarrows M$  over a symplectic manifold  $(M, \omega)$ , with  $t = \text{pr}_1$ ,  $s = \text{pr}_2$  and  $\Omega = \text{pr}_1^* \omega - \text{pr}_2^* \omega$ . Chapter 1 computes its Lie algebroid in Example 1.5. The fibre at the unit  $(x, x)$  is  $\ker T \text{pr}_1 = \{(0, b) : b \in T_x M\}$  identified with  $T_x M$ , and the anchor is the identity. The induced structure on the base is the bracket of  $\omega$  by Example 3.3. With  $Tu(a) = (a, a)$ ,

$$\langle \Psi(0, b), a \rangle = -\Omega((0, b), (a, a)) = -(\omega(0, a) - \omega(b, a)) = \omega(b, a),$$

so  $\Psi(b) = \omega^\flat(b)$ . The theorem says that  $\omega^\flat$  is an isomorphism from the tangent algebroid onto the cotangent algebroid of  $(M, \omega)$ . That is the isomorphism of Example 2.3 read in the opposite direction.

## Chapter 4

# Lie–Poisson Reduction of the Rigid Body

This chapter applies the general framework developed above to the classical free rigid body. The purpose is not to introduce a new reduction theory. Instead, we derive the standard Lie–Poisson reduction and reconstruction formulas directly from the cotangent symplectic groupoid. The Lie–Poisson reduction theorem, as stated in Chapter 13 of Marsden and Ratiu [11], says that a left-invariant Hamiltonian on  $T^*G$  descends to  $\mathfrak{g}^*$  with the minus Lie–Poisson bracket, while the momentum recorded by right translation is conserved. We read this theorem on the cotangent groupoid  $T^*G \rightrightarrows \mathfrak{g}^*$  of Example 3.2.

### 4.1. The cotangent groupoid

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ . We use the cotangent groupoid of Example 3.2 in the left trivialisation  $T^*G \cong G \times \mathfrak{g}^*$  of (3.1.4). Its structure maps are

$$s(g, \mu) = \mu, \quad t(g, \mu) = \text{Ad}_g^* \mu, \quad u(\mu) = (e, \mu),$$

with product  $(h, \text{Ad}_g^* \mu)(g, \mu) = (hg, \mu)$  and inverse  $(g, \mu)^{-1} = (g^{-1}, \text{Ad}_g^* \mu)$ . The symplectic form is  $\Omega = -d\theta$ , where  $\theta$  is the tautological form (3.1.5). On  $\mathfrak{g}^*$  write  $\{\cdot, \cdot\}_+$  for the Lie–Poisson bracket of Definition 2.4 and  $\{\cdot, \cdot\}_-$  for its opposite,

$$\{f_1, f_2\}_\pm(\mu) = \pm \langle \mu, [d_\mu f_1, d_\mu f_2] \rangle, \quad f_1, f_2 \in C^\infty(\mathfrak{g}^*).$$

Here  $d_\mu f \in \mathfrak{g}$  is the differential of Section 2.3.1. It is defined by  $\langle \kappa, d_\mu f \rangle = df_\mu(\kappa)$  for  $\kappa \in \mathfrak{g}^*$ , the derivative of  $f$  at  $\mu$  along  $\kappa$ .

A tangent vector to  $G \times \mathfrak{g}^*$  at  $(g, \mu)$  is a pair  $(v, \lambda)$  with  $v \in T_g G$  and  $\lambda \in \mathfrak{g}^*$ . The Maurer–Cartan form carries the first slot to the identity,

$$\xi = \theta_g^L(v) \in \mathfrak{g}, \quad v = T_e L_g \xi. \quad (4.1.1)$$

We call  $(\xi, \lambda) \in \mathfrak{g} \oplus \mathfrak{g}^*$  the *body components* of  $(v, \lambda)$ . From now on we write tangent vectors in body components. A curve  $\tau \mapsto (g(\tau), \mu(\tau))$  has body velocity  $(\theta_g^L(\dot{g}), \dot{\mu})$ . The source map is the projection to the second factor, and (4.1.1) only changes the first slot. It follows that the differential of the source is

$$Ts(\xi, \lambda) = \lambda. \quad (4.1.2)$$

For  $\xi \in \mathfrak{g}$  let  $\xi^L$  be the left-invariant vector field on  $G$  with  $\xi^L(e) = \xi$ . For  $\lambda \in \mathfrak{g}^*$  write  $\lambda$  also for the constant vector field on the vector space  $\mathfrak{g}^*$ . The product field  $(\xi^L, \lambda)$  on  $G \times \mathfrak{g}^*$  has body components  $(\xi, \lambda)$  at every point. By (3.1.5) the tautological form reads only the first slot,

$$\theta_{(g, \mu)}(\xi, \lambda) = \langle \mu, \xi \rangle. \quad (4.1.3)$$

**Lemma 4.1** (The canonical form in the left trivialisation). *For every  $(g, \mu) \in T^*G$  and all  $(\xi, \lambda), (\eta, \kappa) \in \mathfrak{g} \oplus \mathfrak{g}^*$ ,*

$$\Omega_{(g, \mu)}((\xi, \lambda), (\eta, \kappa)) = \langle \kappa, \xi \rangle - \langle \lambda, \eta \rangle + \langle \mu, [\xi, \eta] \rangle. \quad (4.1.4)$$

*Proof.* The value of  $d\theta$  on two tangent vectors at a point can be computed from any two vector fields that extend them. The product fields  $V = (\xi^L, \lambda)$  and  $W = (\eta^L, \kappa)$  extend  $(\xi, \lambda)$  and  $(\eta, \kappa)$ . For them

$$d\theta(V, W) = V(\theta(W)) - W(\theta(V)) - \theta([V, W]).$$

The body components of  $W$  are  $(\eta, \kappa)$  at every point, so by (4.1.3) the function  $\theta(W)$  takes the value  $\langle \mu, \eta \rangle$  at  $(g, \mu)$ . It does not depend on  $g$ . The integral curve of  $V$  through  $(g, \mu)$  is  $\tau \mapsto (g \exp(\tau\xi), \mu + \tau\lambda)$ . Hence

$$V(\theta(W)) = \left. \frac{d}{d\tau} \right|_{\tau=0} \langle \mu + \tau\lambda, \eta \rangle = \langle \lambda, \eta \rangle.$$

The same argument with the roles exchanged gives  $W(\theta(V)) = \langle \kappa, \xi \rangle$ .

The Lie bracket of two product fields is the product of the brackets of the factors. On  $G$  one has  $[\xi^L, \eta^L] = [\xi, \eta]^L$ . On  $\mathfrak{g}^*$  constant fields commute. Therefore  $[V, W] = ([\xi, \eta]^L, 0)$  and  $\theta([V, W]) = \langle \mu, [\xi, \eta] \rangle$ . Assembling the three terms,

$$d\theta(V, W) = \langle \lambda, \eta \rangle - \langle \kappa, \xi \rangle - \langle \mu, [\xi, \eta] \rangle.$$

Then,  $\Omega = -d\theta$  is (4.1.4). □

**Lemma 4.2.** *Let  $f \in C^\infty(\mathfrak{g}^*)$ . At every point  $(g, \mu)$ , in body components,*

$$X_{s^*f}(g, \mu) = (d_\mu f, \text{ad}_{d_\mu f}^* \mu). \quad (4.1.5)$$

*Proof.* The chain rule gives  $d(s^*f) = df \circ Ts$ . By (4.1.2) and the definition of  $d_\mu f$ , for every  $(\eta, \kappa) \in \mathfrak{g} \oplus \mathfrak{g}^*$

$$d(s^*f)_{(g, \mu)}(\eta, \kappa) = df_\mu(Ts(\eta, \kappa)) = df_\mu(\kappa) = \langle \kappa, d_\mu f \rangle.$$

Let  $(\xi, \lambda)$  be the body components of  $X_{s^*f}(g, \mu)$ . The identity  $\Omega(X_{s^*f}, \cdot) = d(s^*f)$  at  $(g, \mu)$  reads by Lemma 4.1

$$\langle \kappa, \xi \rangle - \langle \lambda, \eta \rangle + \langle \mu, [\xi, \eta] \rangle = \langle \kappa, d_\mu f \rangle$$

for every  $(\eta, \kappa)$ . The choice  $\eta = 0$  gives  $\langle \kappa, \xi \rangle = \langle \kappa, d_\mu f \rangle$  for every  $\kappa$ , so  $\xi = d_\mu f$ . The choice  $\kappa = 0$  then gives  $\langle \lambda, \eta \rangle = \langle \mu, [d_\mu f, \eta] \rangle = \langle \text{ad}_{d_\mu f}^* \mu, \eta \rangle$  for every  $\eta$ , so  $\lambda = \text{ad}_{d_\mu f}^* \mu$ . □

A curve  $\tau \mapsto (g(\tau), \mu(\tau))$  is an integral curve of  $X_{s^*f}$  exactly when its body velocity  $(\theta_g^L(\dot{g}), \dot{\mu})$  equals (4.1.5). By (4.1.1) this is

$$\dot{g} = T_e L_g(d_\mu f), \quad \dot{\mu} = \text{ad}_{d_\mu f}^* \mu. \quad (4.1.6)$$

The first equation moves  $g$  by the left-invariant field of  $d_\mu f$ . The second equation is  $Ts(X_{s^*f})$  by (4.1.2).

Let  $f_1, f_2 \in C^\infty(\mathfrak{g}^*)$  and put  $\xi_i = d_\mu f_i$ . Lemma 4.2 gives  $X_{s^*f_i}(g, \mu) = (\xi_i, \text{ad}_{\xi_i}^* \mu)$ . Put the two vectors into  $\{s^*f_1, s^*f_2\}_\Omega = \Omega(X_{s^*f_1}, X_{s^*f_2})$  and use (4.1.4) and the definition of  $\text{ad}^*$ ,

$$\begin{aligned} \{s^*f_1, s^*f_2\}_\Omega(g, \mu) &= \langle \text{ad}_{\xi_2}^* \mu, \xi_1 \rangle - \langle \text{ad}_{\xi_1}^* \mu, \xi_2 \rangle + \langle \mu, [\xi_1, \xi_2] \rangle \\ &= \langle \mu, [\xi_2, \xi_1] \rangle - \langle \mu, [\xi_1, \xi_2] \rangle + \langle \mu, [\xi_1, \xi_2] \rangle \\ &= -\langle \mu, [\xi_1, \xi_2] \rangle = \{f_1, f_2\}_-(\mu). \end{aligned} \quad (4.1.7)$$

By Theorem 3.1 the source is a Poisson map onto  $(\mathfrak{g}^*, -\pi)$ , where  $\pi$  is the structure for which  $t$  is Poisson. The source is surjective, so  $s^*$  is injective, and (4.1.7) reads  $s^*\{f_1, f_2\}_{-\pi} = s^*\{f_1, f_2\}_-$ . Hence  $-\pi$  is the minus bracket and  $\pi$  is the plus bracket.

Left translation  $L_h$  lifts to  $T^*G$  as the pullback by  $L_{h^{-1}}$ . It carries  $\alpha_g \in T_g^*G$  to  $\alpha_g \circ T_{hg}L_{h^{-1}} \in T_{hg}^*G$ . Since  $L_{h^{-1}} \circ L_{hg} = L_g$ , the left record (3.1.4) of the image is

$$\langle \alpha_g \circ T_{hg}L_{h^{-1}}, T_eL_{hg}\xi \rangle = \langle \alpha_g, T_eL_g\xi \rangle = \langle \mu, \xi \rangle.$$

This shows that in the trivialisation the lift is

$$\widehat{L}_h(g, \mu) = (hg, \mu). \quad (4.1.8)$$

Its orbits are the source fibres  $G \times \{\mu\}$ . A function on  $T^*G$  is invariant under every  $\widehat{L}_h$  exactly when it is a source pullback  $s^*H$ .

**Theorem 4.1** (Lie–Poisson reduction). *Let  $F \in C^\infty(T^*G)$  be invariant under (4.1.8) and put  $H = F \circ u$ . Then  $F = s^*H$ , and wherever it is defined the flow  $\widehat{\varphi}_\tau$  of  $X_F$  satisfies*

$$t \circ \widehat{\varphi}_\tau = t, \quad s \circ \widehat{\varphi}_\tau = \varphi_\tau^- \circ s,$$

where  $\varphi_\tau^-$  is the Hamiltonian flow of  $H$  on  $(\mathfrak{g}^*, \{\cdot, \cdot\}_-)$ .

*Proof.* An invariant function is constant on each source fibre, so  $F(g, \mu) = F(e, \mu) = H(\mu)$  and  $F = s^*H$ . Corollary 3.1(ii) gives  $t \circ \widehat{\varphi}_\tau = t$  and  $s \circ \widehat{\varphi}_\tau = \varphi_{-\tau} \circ s$  for the flow  $\varphi_\tau$  of  $H$  on  $(\mathfrak{g}^*, \pi)$ . The structure  $\pi$  is the plus bracket, and  $\{\cdot, H\}_- = -\{\cdot, H\}_+$ , so  $\varphi_{-\tau} = \varphi_\tau^-$ .  $\square$

## 4.2. The free rigid body

Take  $G = \mathrm{SO}(3)$  with the identifications of Section 2.3.4. The hat map identifies  $\mathfrak{so}(3)$  with  $\mathbb{R}^3$  and the dot product identifies  $\mathfrak{so}(3)^*$  with  $\mathbb{R}^3$ . Then

$$[\xi, \eta] = \xi \times \eta, \quad \mathrm{Ad}_R \xi = R\xi, \quad \mathrm{Ad}_R^* \mu = R\mu, \quad \mathrm{ad}_\xi^* \mu = \mu \times \xi, \quad \mathrm{d}_\mu f = \nabla f.$$

The group  $\mathrm{SO}(3)$  is connected. Its coadjoint orbits are the spheres  $|\mu| = \text{const}$  and the origin. By Theorem 2.2 these are the symplectic leaves of  $\mathfrak{so}(3)^*$ . The smooth Casimirs are precisely the  $\mathrm{SO}(3)$ -invariant smooth functions; equivalently, they can be written as smooth functions of  $|\mu|^2$ .

An arrow  $(R, \mu) \in T^*\mathrm{SO}(3)$  is an attitude  $R$  and a vector  $\mu \in \mathbb{R}^3$ . Its source is  $\mu$ . Its target is

$$J = R\mu = t(R, \mu).$$

For the rigid body  $\mu$  is the angular momentum in the body frame and  $J$  is the angular momentum in the space frame. This is the standard identification of the left and right records of a covector on  $\mathrm{SO}(3)$  with the body and spatial momenta [11, Chap. 15]. Let  $\tau \mapsto (R(\tau), \mu(\tau))$  be a curve in  $T^*\mathrm{SO}(3)$ . Its body velocity is  $(\xi, \lambda)$  with  $\xi = \theta_R^L(\dot{R})$  and  $\lambda = \dot{\mu}$ . On the matrix group  $\mathrm{SO}(3)$  left translation by  $R$  is multiplication by the matrix  $R$ , so  $T_eL_R \hat{\xi} = R\hat{\xi}$  and

$$\theta_R^L(\dot{R}) = R^{-1}\dot{R} = R^\top \dot{R}.$$

Differentiating  $R^\top R = 1$  gives  $\dot{R}^\top R + R^\top \dot{R} = 0$ , so  $R^\top \dot{R}$  is skew and  $R^\top \dot{R} = \hat{\xi}$  for a unique  $\xi \in \mathbb{R}^3$ . Lemma 4.1 becomes

$$\Omega_{(R, \mu)}((\xi, \lambda), (\eta, \kappa)) = \kappa \cdot \xi - \lambda \cdot \eta + \mu \cdot (\xi \times \eta),$$

and (4.1.7) becomes

$$\{s^*f_1, s^*f_2\}_\Omega = -\mu \cdot (\nabla f_1 \times \nabla f_2),$$

the minus bracket of Section 2.3.4 in vector form.

The kinetic energy of the body is

$$H(\mu) = \frac{1}{2} \mu \cdot I^{-1} \mu,$$

a function of the body momentum alone, because the inertia tensor  $I$  is constant in the body frame. By (4.1.8) the function  $s^*H$  is invariant under  $\widehat{L}_Q(R, \mu) = (QR, \mu)$ , and Theorem 4.1 applies with  $F = s^*H$ . The gradient  $\nabla H = I^{-1}\mu = \omega$  is the body angular velocity. Lemma 4.2 gives

$$X_{s^*H} = (\omega, \operatorname{ad}_\omega^* \mu) = (\omega, \mu \times \omega).$$

With  $T_e L_R \hat{\omega} = R\hat{\omega}$  the equations (4.1.6) with  $f = H$  read

$$\dot{R} = R\hat{\omega}, \quad \dot{\mu} = \mu \times \omega, \quad \omega = I^{-1}\mu. \quad (4.2.1)$$

The first equation is the kinematic relation between attitude and body angular velocity. The second is Euler's equation (2.3.2). The two equations are the two slots of the one field  $X_{s^*H}$  on  $T^*\mathrm{SO}(3)$ . Energy is conserved because it is the Hamiltonian. The smooth Casimir  $C(\mu) = \frac{1}{2}|\mu|^2$  is conserved. Consequently,  $|\mu|$  is conserved as well. Trajectories of  $\mu$  lie in the intersections of the spheres  $|\mu| = \text{const}$  with the ellipsoids  $H = \text{const}$ .

A solution of (4.2.1) keeps  $\mu$  on a sphere and  $R$  in the compact group  $\mathrm{SO}(3)$ . It stays in a compact set, so it exists for all  $\tau$ . Start at the unit  $(1, \mu_0)$ . By Theorem 4.1 the target is constant,  $R(\tau)\mu(\tau) = \mu_0$ , so

$$\mu(\tau) = R(\tau)^\top \mu_0.$$

The arrow  $(R(\tau), \mu(\tau))$  lies in the bisection  $B_\tau$  of Proposition 3.7 and has target  $\mu_0$ . Consequently  $\operatorname{Ad}_{R(\tau)}^* \mu(\tau) = \mu_0$  is (3.4.3) for this arrow.

Write  $R(\tau, \mu_0)$  for the attitude at time  $\tau$  of the solution that starts at  $(1, \mu_0)$ . The flow of  $X_{s^*H}$  is defined for all  $\tau$  on the whole unit section, so Proposition 3.7 applies for every  $\tau$ . The bisection  $B_\tau$  consists of the arrows  $(R(\tau, \mu_0), R(\tau, \mu_0)^\top \mu_0)$ . Inversion exchanges  $s$  with  $t$  and carries Lagrangian submanifolds to Lagrangian submanifolds by Lemma 3.1(iii), so  $\iota(B_\tau)$  is again a Lagrangian bisection. Its arrows are  $(R(\tau, \mu_0)^\top, \mu_0)$ , so in the description of Example 3.5 it is the map  $\mu_0 \mapsto R(\tau, \mu_0)^\top$ . Its base map (3.4.2) is  $\mu_0 \mapsto R(\tau, \mu_0)^\top \mu_0 = \mu(\tau)$ , the time  $\tau$  flow of Euler's equation.

Write  $\ell_i(\nu) = \nu_i$  for the linear coordinates on  $\mathfrak{so}(3)^*$ . On  $T^*\mathrm{SO}(3)$  put  $\mu_i = s^*\ell_i$  and  $J_i = t^*\ell_i$ , the body and space components of the momentum. By Section 2.3.1 the plus bracket of linear functions is  $\{\ell_i, \ell_j\}_+ = \ell_{[e_i, e_j]} = \ell_{e_i \times e_j} = \varepsilon_{ijk}\ell_k$ . The source is Poisson onto the minus bracket by (4.1.7), so

$$\{\mu_i, \mu_j\}_\Omega = s^*\{\ell_i, \ell_j\}_- = -\varepsilon_{ijk}\mu_k.$$

The target is Poisson onto the plus bracket by Section 4.1, so

$$\{J_i, J_j\}_\Omega = t^*\{\ell_i, \ell_j\}_+ = \varepsilon_{ijk}J_k.$$

Source and target pullbacks Poisson-commute by Lemma 3.2(iii), so  $\{J_i, \mu_j\}_\Omega = 0$ . Together

$$\{J_i, J_j\}_\Omega = \varepsilon_{ijk}J_k, \quad \{\mu_i, \mu_j\}_\Omega = -\varepsilon_{ijk}\mu_k, \quad \{J_i, \mu_j\}_\Omega = 0. \quad (4.2.2)$$

# Conclusion

This Bachelor thesis has presented a coherent geometric route from Lie groupoids and Lie algebroids to Poisson manifolds and symplectic groupoids, and has used this route to describe Lie–Poisson reduction of the free rigid body. The main objective was not to introduce a new theory, but to explain how several standard constructions of geometric mechanics fit together and how symplectic groupoids provide a global counterpart to the infinitesimal structures arising in Poisson geometry.

The first part of the thesis introduced Lie groupoids as a natural extension of Lie groups when one wants to encode local or partial symmetries rather than only globally defined transformations. Their infinitesimal counterparts, Lie algebroids, were then obtained by linearising the groupoid structure along the unit section. This makes precise the analogy between Lie groups and Lie algebras while allowing the infinitesimal data to vary over a manifold. In particular, the anchor map records how infinitesimal symmetries act on the base, whereas the bracket on sections contains the corresponding algebraic information.

The second part moved to Poisson geometry. A Poisson manifold allows Hamiltonian dynamics even when the phase space is not globally symplectic. Its characteristic distribution decomposes the manifold into symplectic leaves, and Casimir functions are constant along every Hamiltonian trajectory. The Lie–Poisson structure on  $\mathfrak{g}^*$  gives the basic example relevant to reduction by symmetry. Its symplectic leaves are the connected components of coadjoint orbits. The construction of the cotangent Lie algebroid  $T^*M$  then provides the infinitesimal object naturally associated with a Poisson manifold.

Symplectic groupoids bring these two viewpoints together. The multiplicativity of the symplectic form implies strong structural properties: the unit section is Lagrangian, inversion is anti-symplectic, and the source and target maps induce opposite Poisson structures on the base. Moreover, the Lie algebroid of a symplectic groupoid is canonically identified with the cotangent Lie algebroid of its Poisson base. Thus, a symplectic groupoid may be regarded as a global integration of the infinitesimal Poisson data. Lagrangian bisections provide, in addition, a natural global description of Poisson transformations; for the pair groupoid they reduce to the usual canonical transformations of symplectic mechanics.

The cotangent groupoid  $T^*G \rightrightarrows \mathfrak{g}^*$  illustrates these ideas concretely. In left trivialisation its source and target maps encode two momentum descriptions of the same covector. For  $G = \mathrm{SO}(3)$ , these become the body and spatial angular momenta of the rigid body. A left-invariant Hamiltonian on  $T^*G$  descends through the source map to the minus Lie–Poisson bracket on  $\mathfrak{g}^*$ . For the free rigid body this reduction yields Euler’s equations, while the target map expresses conservation of the spatial angular momentum. Reconstruction recovers the attitude variable from the reduced motion, and the two signs in the Poisson brackets of body and spatial momenta arise directly from the source–target structure of the symplectic groupoid.

The rigid-body example therefore shows the main point of the thesis. Symplectic groupoids organise the standard formulations of Hamiltonian mechanics into a single geometric pic-

ture in which unreduced symplectic dynamics, reduced Poisson dynamics, infinitesimal Lie-algebroid data and reconstruction are related by the same global structure. This viewpoint is useful precisely because it separates what is intrinsic to the Poisson base from what depends on a chosen global integration.

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